
∞ LEARNING WHAT TO REMEMBER: PREDICTIVE ADMISSION FOR BOUNDED REASONING-TIME MEMORY

MMLA FOCUS PAPER

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Project: github.com/MMLA-org/mmla-memory

ABSTRACT

A bounded memory cannot keep every completed observation. It must preserve the current state, commit one complete target-conditioned row, or decline the write when evidence is insufficient. We formalize this decision before training a controller. A completed segment is eventized; event j constructs and scores actions against the branch state produced by events $1:j-1$. Training-only groups of alternative futures share a causal prefix and price every feasible action, including exact NULL, through action-conditioned replay. Deployment sees only a typed causal information set.

We define horizon risk with task loss, stale leakage, collateral damage, write and NULL penalties, and resource charges. We distinguish an expected-risk oracle from a current-risk comparator and a future-blind value model, and require simultaneous confidence control before declaring an oracle gap. Under explicit exchangeability, support, bounded-loss, replay, and evaluator assumptions, we prove weighted concentration, stable action recovery, uncertainty-aware admission relative to NULL, event-recursive consistency, a 2ϵ value-to-regret bound, branch-shift and evaluator-error bounds, and non-identification under missing action support. Counterexamples expose post-treatment leakage, synthetic-future bias, hidden confounding, and sub-resolution gaps. The full ledger makes the $|\mathcal{A}| \times K \times H$ replay multiplier explicit. No new experiment, measured oracle gap, or learned admission policy is reported; the same-checkpoint semantic interface remains unqualified and predictive overwrite remains closed.

1 Introduction

1.1 Admission is a decision about authority

A transcript may retain a sentence, a retriever may return it, and a recurrent state may compress it. None of those operations alone decides whether the sentence should replace a current authoritative fact. Under capacity pressure, a write has at least three possible costs: it may admit false or stale content, displace a useful row, or spend resources without improving any later decision. Refusing a write has a different cost: the state may remain stale or omit information that would have helped a later task. Predictive memory admission is the decision problem created by those asymmetric consequences.

DEFINITION

Definition 1.1 (Predictive memory admission). Predictive memory admission is the selection of one action from a finite feasible set containing exact NULL and zero or more complete target-conditioned row commits, using only information available at the decision boundary, with the objective and evidence gate fixed before evaluation futures are opened.

The word *predictive* refers to conditional future consequence, not access to a realized future at deployment. A training teacher may use later outcomes to estimate action risk. A deployed value model may use only the causal state. The difference is structural and will be expressed as a filtration constraint rather than a promise in prose.

1.2 Evidence boundary

This manuscript inherits architecture and identity from the public MMLA V3 manuscript and its technical-report evidence record [13]. Their evidence boundary is unchanged. The registered same-checkpoint semantic interfaces did not qualify. PM-12 qualified zero of nine jobs. A subsequent fitted-population diagnostic reported learned content exact on 0/73,728

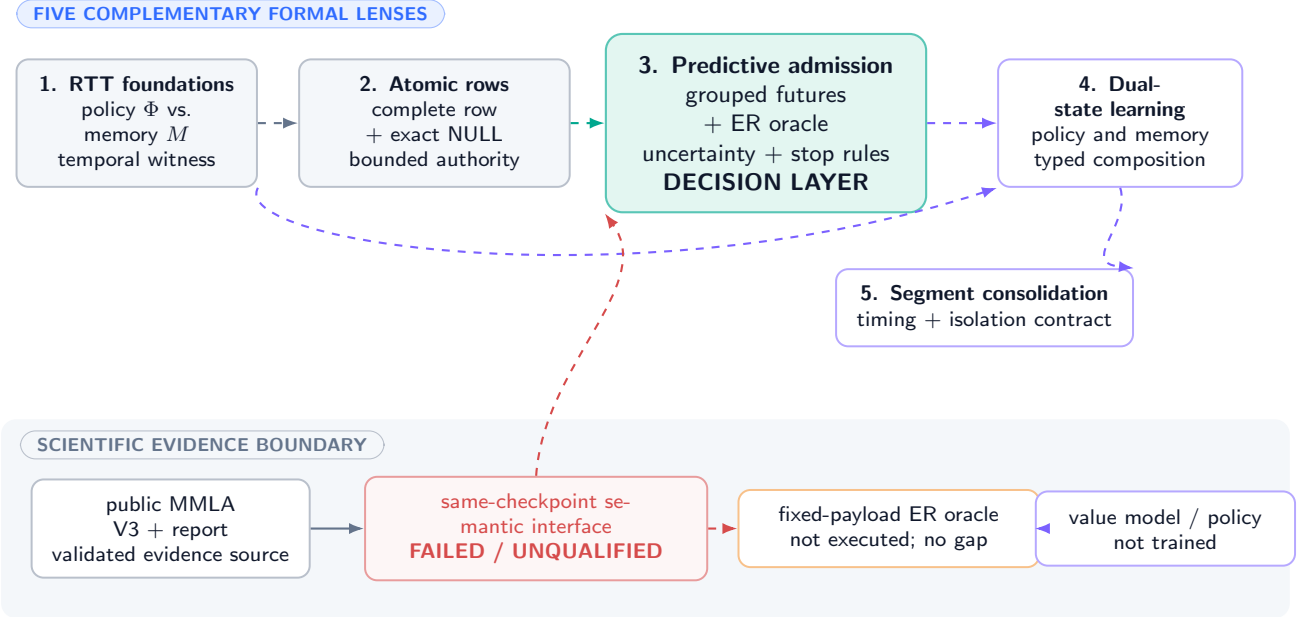
probes both with learned routing and when the target row was supplied mechanically; only the complete mechanical oracle was perfect. The result is a fit/support failure for the tested pure latent-row interface, not a rejection of every possible memory representation.

UNVALIDATED Consequently, the empirical expected-risk oracle in this paper has not been run, no positive oracle gap exists in the admitted record, and no future-blind value model or admission policy has been learned. Predictive overwrite remains closed by gate. Every numerical example below is identified as illustrative; every experimental protocol is future work.

The companion focus papers provide terminology and an external row contract only. The RTT foundation separates mutable policy state from memory state [14]. Atomic Memory Rows supplies complete-row, trusted-assembly, version, protection, and exact-NUL semantics [12]. Neither formal dependency supplies empirical evidence for admission.

MMLA focus-paper family: predictive admission at a closed gate

conceptual dependencies organize interfaces; empirical status never transfers across an arrow



Coral marks one bounded inherited stop. This paper defines unopened gates; it reports no oracle gap, learned content, route, value, or policy.

Figure 1: **Predictive admission in the five-paper conceptual map.** The coral stop at the inherited semantic-interface gate blocks any oracle, controller, or overwrite-success claim. Formal dependency arrows do not transfer empirical status.

1.3 Contributions

This paper owns the following questions.

1. It types the causal state, completed segment, event sequence, bounded resident memory, target-conditioned candidate, action set, grouped future, horizon loss, resource ledger, and deployable information set.
2. It gives event-indexed teacher semantics. Event j reads the branch state produced by earlier committed events; it never acts on a whole-segment candidate computed against an obsolete prefix.
3. It defines grouped-future sampling and action-conditioned replay, including branch weights, support, synthetic-future bias, stale leakage, collateral damage, write/NULL penalties, and resources.
4. It separates an empirical expected-risk oracle, a current-risk comparator, exact NULL, and a future-blind value model.
5. It supplies simultaneous concentration and conservative uncertainty rules, identification and impossibility results, metrics, leakage audits, full costs, stop rules, and theorem-to-implementation obligations.

This paper does not specify a row serializer, trusted assembler implementation, semantic reader, event extractor, archive policy, policy-state RTT update, or complete MMLA system. It does not run a training job, choose a checkpoint, or estimate an oracle gap. Definitions, conditional proofs, prescriptions, and planned tests are marked locally; none constitutes implementation evidence.

1.4 Central falsifiable question

For a causal prefix state $\tilde{S}$, let $R(a \mid \tilde{S})$ be the conditional horizon risk of feasible action a , let a^{cur} be a preregistered current-risk action, and let 0 denote NULL. The first scientific question is

$$G(\tilde{S}) = R(a^{\text{cur}} \mid \tilde{S}) - \min_{a \in \mathcal{A}(\tilde{S})} R(a \mid \tilde{S}). \quad (1)$$

The relevant population claim is not that G can be positive in a hand-built example. It is that a frozen empirical procedure yields a stable, uncertainty-controlled improvement exceeding a prespecified materiality margin on fresh prefix groups and required seeds. If it does not, the value model remains closed.

CONJECTURED Some domains may contain predictable future-use structure for which a bounded authoritative memory benefits from selective admission. This conjecture can fail because the action set is poor, futures are not identified, the gap is too small, the value is not predictable from causal history, long-run pollution dominates, or total-system cost erases the benefit.

2 Typed Causal State and Deployable Information

2.1 Probability space and completed segments

Fix a probability space $(\Omega, \mathcal{B}, \mathbb{P})$ and a discrete event clock with filtration $(\mathcal{F}_t)_{t \geq 0}$. Stream instance n has identity $I_n \in \mathcal{I}$, observations $X_{n,t} \in \mathcal{X}$, and a completed segment

$$B_n = (X_{n,b_{n-1}+1}, \dots, X_{n,b_n}) \quad (2)$$

that becomes available only at stopping time b_n . Prediction at or before b_n cannot use a memory mutation produced after B_n completes. A bounded eventizer maps the completed segment and prior state to an ordered list

$$E_n = \text{Eventize}(B_n, M_{n-1}; \xi_n) = (e_{n,1}, \dots, e_{n,J_n}), \quad 0 \leq J_n \leq J_{\max}, \quad (3)$$

where ξ_n is recorded eventizer randomness. Eventization is itself a costed, unvalidated component; the theory conditions on its declared output.

Object	Space	Meaning	Boundary
I_n	$\mathcal{I}$	Prefix-group / stream identity	Split and receipt unit
B_n	bounded sequence	Completed observed segment	No unobserved continuation
$e_{n,j}$	$\mathcal{E}$	Event j in the ordered segment list	Processed only after events $< j$
$M_{n,j}$	$\mathcal{M} = \mathcal{R}^S$	S complete authoritative rows after event j	Fixed resident capacity
$c_{n,j,a}$	$\mathcal{R} \cup \{\perp\}$	Complete row candidate for target a	Built against current branch prefix
$A_{n,j}$	finite subset	Feasible commits plus NULL	Hard mask precedes value scoring
$F_{n,k}$	$\mathcal{F}$	Future branch k for the same prefix	Training/evaluation only
H	$\mathbb{N}$	Replay horizon	Fixed before outcomes open
L	$[0, L_{\max}]$	Registered aggregate action loss	Bounded for finite-sample results
$\mathfrak{R}$	$\mathbb{R}_+^d$	Resource ledger	Logical and measured coordinates
$\tilde{S}_{n,j}$	$\mathcal{S}$	Deployable causal information	No future-bearing field

Table 1: Typed objects. Physical co-location does not merge their causal roles.

2.2 Bounded authoritative state

Let

$$M_{n,j} = (r_{n,j,1}, \dots, r_{n,j,S}) \in \mathcal{R}^S \quad (4)$$

be the complete resident state. The row space $\mathcal{R}$ and slot count S are finite-width under the external contract. This paper assumes a trusted validator can distinguish a complete committable row from $\perp$; it does not assume that the row is factually true or semantically usable.

For event j , candidate construction for target $a \in \{1, \dots, S\}$ is the partial map

$$c_{n,j,a} = \text{Construct}(\tilde{S}_{n,j}, e_{n,j}, a, r_{n,j-1,a}, M_{n,j-1}; \zeta_{n,j,a}) \in \mathcal{R} \cup \{\perp\}, \quad (5)$$

with fixed or recorded constructor randomness $\zeta_{n,j,a}$. Target conditioning is essential because version, protection, provenance, collision, eviction, tenant, and rollback fields can depend on the old target row. A shared semantic payload may be a restricted causal probe, but the authoritative action still requires a complete target-specific assembly.

The feasible set is

$$\mathcal{A}_{n,j} = \{0\} \cup \{a \in \{1, \dots, S\} : c_{n,j,a} \neq \perp, \text{Valid}(c_{n,j,a}; M_{n,j-1}) = 1\}, \quad (6)$$

where action 0 denotes exact NULL. Authorization, protection, schema, version, tenant, and snapshot checks are hard predicates. A learned score cannot restore an excluded action.

2.3 Deployable information set

Let $t_{n,j}^-$ be the instant before the event- j decision. The deployable sigma-field is

$$\mathcal{D}_{n,j} = \sigma\left(I_n, X_{n,\leq b_n}, B_n, e_{n,\leq j}, M_{n,j-1}, \{c_{n,j,a}\}_{a \in \mathcal{A}_{n,j} \setminus \{0\}}, \mathfrak{R}_{n,j}^{\text{remaining}}, \Xi_{n,j}\right) \subseteq \mathcal{F}_{t_{n,j}^-}, \quad (7)$$

where $\Xi_{n,j}$ contains declared causal metadata, random state, and calibrated uncertainty state. The bounded summary is a measurable map

$$\tilde{S}_{n,j} = \Psi_{\text{dep}}(\mathcal{D}_{n,j}) \in \mathcal{S}. \quad (8)$$

DEFINITION

Definition 2.1 (Future-blind deployment). A deployed scorer or decision rule is future blind at event (n, j) if its output is $\mathcal{D}_{n,j}$ -measurable and its executable input graph has no continuation, future token, future answer, future evaluator output, branch identifier, or statistic computed from any of those values.

DESIGN PRESCRIPTION Declaring an input “history only” is insufficient. A conformance audit must enumerate every feature producer, cached tensor, RNG state, retrieved artifact, label join, and preprocessing path that can reach the deployment call.

2.4 Core assumptions

Assumption 2.2 (Causal timing). Every event decision is $\mathcal{D}_{n,j}$ -measurable, and the committed state becomes readable only after the decision time. No future branch affects eventization, candidate construction, feasibility, or deployment features.

Assumption 2.3 (External row contract). For every accepted action, the constructor and trusted assembler return one complete row bound to the current snapshot; NULL is exact identity; hard validation is deterministic for fixed inputs; and one event changes at most its selected target.

Assumption 2.4 (Finite action and loss). $1 \leq |\mathcal{A}_{n,j}| < \infty$ and every registered horizon loss lies in $[0, L_{\max}]$. If a natural loss is unbounded, the protocol fixes clipping or a different tail model before evaluation.

These assumptions support later results but are not asserted to hold in the frozen system. In particular, Assumption 2.3’s semantic precondition is presently unvalidated.

3 Event-Indexed Candidates, Actions, and Teacher Timing

3.1 Deployed recursion

Set $M_{n,0} = M_{n-1}$. After constructing $\mathcal{A}_{n,j}$ against $M_{n,j-1}$, a deployed decision rule selects $A_{n,j} \in \mathcal{A}_{n,j}$ and commits

$$M_{n,j} = T_{n,j}(M_{n,j-1}, A_{n,j}) = \begin{cases} M_{n,j-1}, & A_{n,j} = 0, \\ \text{Commit}(M_{n,j-1}, A_{n,j}, c_{n,j,A_{n,j}}), & A_{n,j} \neq 0. \end{cases} \quad (9)$$

The segment output is $M_n = M_{n,J_n}$. Later events see earlier committed events through this state and no other within-segment authority.

PROVED UNDER STATED ASSUMPTIONS

Proposition 3.1 (Event-timing consistency). *Under Assumptions 2.2 and 2.3, $M_{n,j}$ is measurable with respect to information available immediately after event j , and every candidate used at event $j + 1$ is a function of $M_{n,j}$ rather than an obsolete pre-segment state.*

Proof. The base state $M_{n,0}$ is available before the first decision. Given an available $M_{n,j-1}$, eventization output $e_{n,j}$, construction inputs, hard validation, action selection, and commit are causal by Assumption 2.2; hence $M_{n,j}$ is available after the event. Equation 5 then uses that exact state for event $j + 1$. Induction over j proves the claim. $\square$

The proposition prohibits a convenient but invalid shortcut: construct all segment candidates against $M_{n,0}$, choose several actions, and describe the result as event-recursive. Earlier commits can change target versions, feasibility, collision status, retained content, and later losses.

3.2 Branch-prefix recursion for a teacher

To price counterfactual action sequences, let $\mathbf{a}_{1:j} = (a_1, \dots, a_j)$. Each branch has its own state

$$M_{n,0}^{()} = M_{n-1}, \quad M_{n,j}^{(\mathbf{a}_{1:j})} = T_{n,j}^{(\mathbf{a}_{<j})} \left(M_{n,j-1}^{(\mathbf{a}_{<j})}, a_j \right), \quad (10)$$

where event summary, candidate set, and feasibility at depth j are recomputed against the branch prefix $M_{n,j-1}^{(\mathbf{a}_{<j})}$. The teacher may use one of three distinct protocols:

1. **Exact dynamic program:** enumerate every feasible branch prefix and use a future-blind continuation policy optimized outside the future expectation.
2. **Frozen rollout:** after the scored action, follow one preregistered future-blind continuation policy for later segment events.
3. **Single-event intervention:** set $J_n = 1$ or forbid later writes in the scoring horizon. This is the cleanest first causal test.

Mixing these protocols is not allowed. In particular, choosing later actions separately for each revealed future is a hindsight policy and is not a deployable cost-to-go.

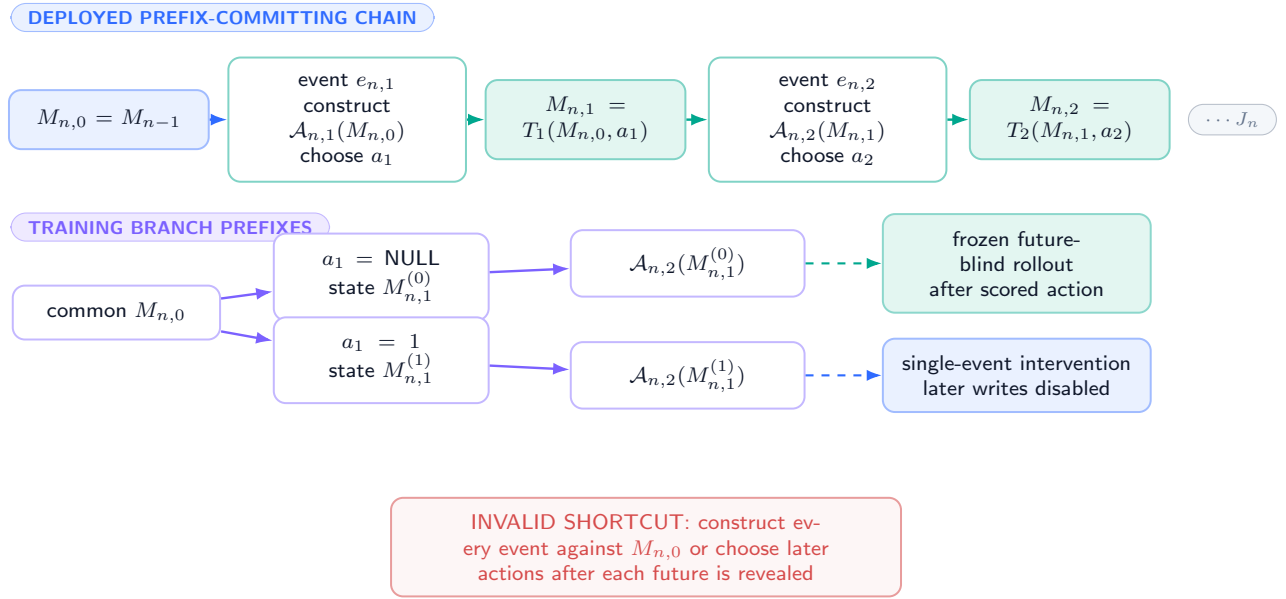
For a sequence policy π whose event- q action is measurable with respect to the branch state before event q , define

$$Q_{n,j}^T(a \mid \tilde{S}_{n,j}^{(\mathbf{a}_{<j})}) = \inf_{\pi \in \Pi_{j+1:J_n}^{\text{fb}}} \mathbb{E} \left[L_H((\mathbf{a}_{<j}, a, A_{j+1:J_n}^\pi), F) \mid \tilde{S}_{n,j}^{(\mathbf{a}_{<j})} \right]. \quad (11)$$

The infimum is outside the expectation over F . Moving it inside would let the teacher observe the realized future before selecting later events.

Event-indexed action recursion: every candidate reads its actual branch prefix

one segment may contain several events; exact, frozen-rollout, and single-event teachers are different estimands



Exact dynamic programming enumerates supported prefix-action paths. Its infimum is outside the future expectation; moving it inside creates hindsight.

Figure 2: **Event-indexed recursion (DEFINITION)**. Candidate/action sets at event j are constructed from the branch state after earlier commits. Exact, frozen-rollout, and single-event teachers are distinct. No whole-segment action may reuse candidates from an obsolete prefix.

3.3 No conflated segment action

PROVED UNDER STATED ASSUMPTIONS

Theorem 3.2 (Sequential composition and support). *Under Assumption 2.3, each legal event changes at most one resident row. Across a segment of J_n events, the final state can differ from M_{n-1} on at most J_n distinct slots. A later NULL leaves earlier committed prefixes unchanged.*

Proof. At one event, NULL changes zero rows and a non-NUL action replaces exactly its selected row. The union of target slots over J_n events therefore has size at most J_n , and every row outside that union remains unchanged. Equation 9 is prefix committing, so an identity transition at event j maps $M_{n,j-1}$ to itself rather than to M_{n-1} . $\square$

This theorem establishes a mechanical support bound only. It does not show that a one-row change is semantically local, that a segment is all-or-nothing, or that a teacher can safely ignore later events.

3.4 Required timing receipt

DESIGN PRESCRIPTION Every future implementation must record, per event and per counterfactual branch: prefix-state digest, event digest, constructor/assembler version, candidate digests, feasibility mask, action, commit or NULL result, successor digest, future-group identifier, rollout protocol, and resource increments. A candidate whose recorded predecessor differs from the replay prefix is invalid even if its bytes parse.

4 Grouped Alternative Futures and Sampling Assumptions

4.1 Physical prefix groups

Fix one event decision state $\tilde{S}_g$ and its immutable pre-action memory snapshot M_g . A grouped-future record is

$$\mathcal{G}_g = \left(g, \tilde{S}_g, M_g, \mathcal{A}_g, \{c_{g,a}\}_{a \in \mathcal{A}_g \setminus \{0\}}, \{(F_{g,k}, w_{g,k})\}_{k=1}^{K_g} \right), \quad (12)$$

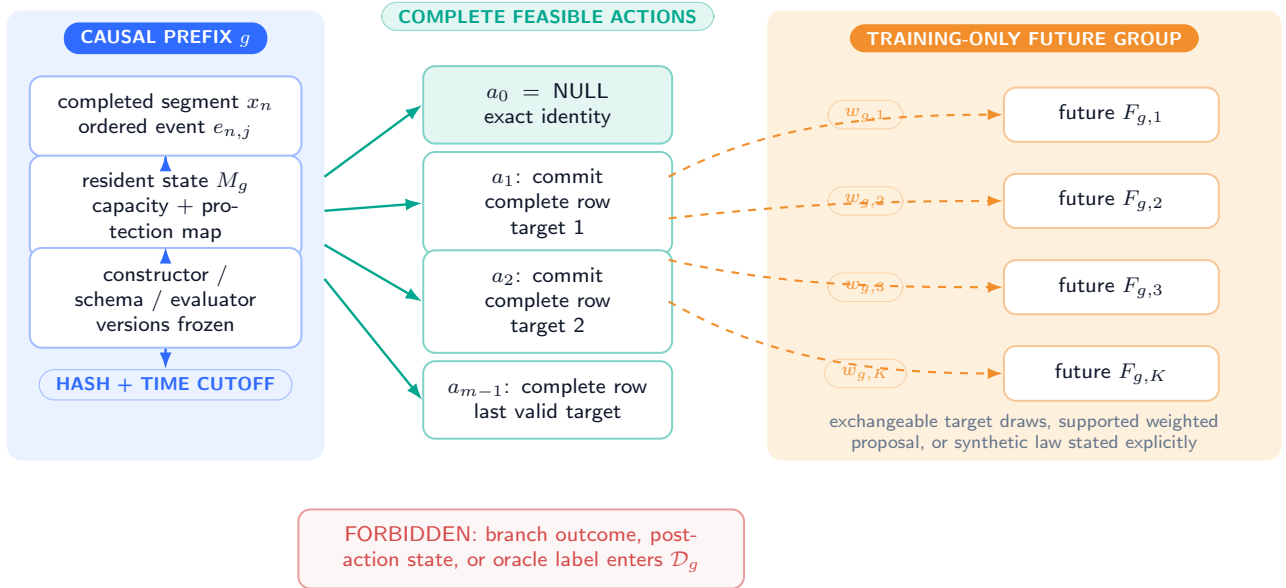
where every branch shares exactly the same causal prefix, candidates, and pre-action snapshot; $w_{g,k} \geq 0$ and $\sum_k w_{g,k} = 1$. The group ID, not a future branch, is the split unit.

DEFINITION

Definition 4.1 (Grouped future). A grouped future is a continuation sampled or constructed for one fixed causal prefix state. “Similar prompts,” paraphrases with different hidden states, or branches joined only after answer labels are known are not grouped futures without an explicit causal equivalence argument.

One causal prefix, a complete action set, and grouped alternative futures

training may expose continuations only after actions and candidate bytes are frozen; deployment sees the prefix panel only



Group identity binds prefix, every action, every future, weights, horizon, loss, RNG streams, and provenance. Split the physical group only as a unit.

Figure 3: **Causal prefix and grouped futures.** Dashed amber branches are training/evaluation-only continuations from one frozen prefix. Solid navy and teal paths are deployable. Future tokens may price actions but no future-derived field crosses the deployment boundary.

4.2 Sampling models

Let P_g be the intended deployment continuation distribution conditional on $\tilde{S}_g$. Three data regimes must be distinguished.

Regime	Identifying requirement	Mandatory limitation
Repeated real continuations	Branches are conditionally exchangeable draws from P_g or a declared finite population	Repeated collection may alter users or environment
Controlled generator	Generator seed and branch law are fixed before action outcomes; support covers the declared target population	Identifies the generator distribution, not natural deployment
Model-generated futures	Generator is frozen and causally separated from action scoring; calibration compares its branch law with fresh real continuations	Self-consistency and mode collapse may bias values
Logged observational futures	Every evaluated action has positive logging support, propensities and confounders are recorded, and a valid off-policy estimator is used	Missing support cannot be repaired by modeling alone

Table 2: Grouped-future regimes. Synthetic and model-generated futures require an explicit target-distribution boundary.

Assumption 4.2 (Conditional exchangeability). For a designed replay group, $(F_{g,1}, \dots, F_{g,K_g})$ are conditionally independent draws from a declared branch law $Q_g(\cdot \mid \tilde{S}_g)$, or the estimator and concentration analysis are replaced by a valid dependence-aware procedure.

Assumption 4.3 (Action support and consistency). Every action whose risk is estimated is feasible at the common snapshot and has a well-defined potential replay outcome. The replay engine returns that outcome when the corresponding action is applied. If logged data rather than full replay are used, the logging propensity is positive wherever the target action has positive probability.

Assumption 4.4 (Exogenous or interventional future). Either the future evaluation sequence is conditionally exogenous to the memory action, so the same branch can be replayed under all actions, or the environment supplies valid action-conditioned counterfactual futures. Reusing one observed future under all actions is not valid when the action changes the future-generating process.

These are causal-identification assumptions in the potential-outcome sense [11, 10]. They are stricter than ordinary predictive train/test separation.

4.3 Branch weights and effective sample size

Weights may encode known branch probabilities, stratified sampling, or a fixed finite-population target. They must not be tuned to increase the preferred action’s advantage. Define

$$K_{\text{eff},g} = \frac{1}{\sum_{k=1}^{K_g} w_{g,k}^2}, \quad (13)$$

which equals K_g for uniform weights and approaches one when a single branch dominates. Reporting only K_g can therefore overstate information.

If branches are sampled under proposal law Q_g but target risk uses P_g , importance weights require $P_g \ll Q_g$ and bounded or otherwise controlled density ratios. Doubly robust off-policy estimators can combine a reward model with propensities [3]; they do not solve zero support or hidden confounding.

4.4 Synthetic-future bias

For any action loss bounded in $[0, L_{\max}]$, define total variation as

$$\text{TV}(P, Q) = \sup_A |P(A) - Q(A)|. \quad (14)$$

Then the risk under a synthetic branch law can differ from deployment risk even with infinitely many synthetic samples. Section 8 proves the bound

$$|R_P(a) - R_Q(a)| \leq L_{\max} \text{TV}(P, Q). \quad (15)$$

The sampling error shrinks with more branches; the distributional bias does not. A model-generated teacher is therefore not an oracle for the real future unless the target relation is independently justified.

4.5 Split obligations

DESIGN PRESCRIPTION Fitting, calibration, and final evaluation must be disjoint at the prefix-family or generating-process level. All branches from one group remain together. Near-duplicate prefixes, shared answer templates, generator IDs, and deterministic seed descendants are audited before labels are opened. Final-only means one frozen model, threshold, horizon, loss vector, and confidence procedure are read once on the held-out groups; a failed seed is never replaced.

5 Action-Conditioned Replay and Horizon Loss

5.1 One immutable snapshot, every feasible action

For group g , action $a \in \mathcal{A}_g$, and future branch $F_{g,k}$, replay begins from the same immutable pre-action snapshot M_g . Define

$$M_g^{(a)} = \begin{cases} M_g, & a = 0, \\ \text{Commit}(M_g, a, c_{g,a}), & a \neq 0. \end{cases} \quad (16)$$

The continuation is then executed under $M_g^{(a)}$. A cached hidden trace computed before the action and merely rescored afterward is not action-conditioned replay when memory can affect future hidden-state evolution.

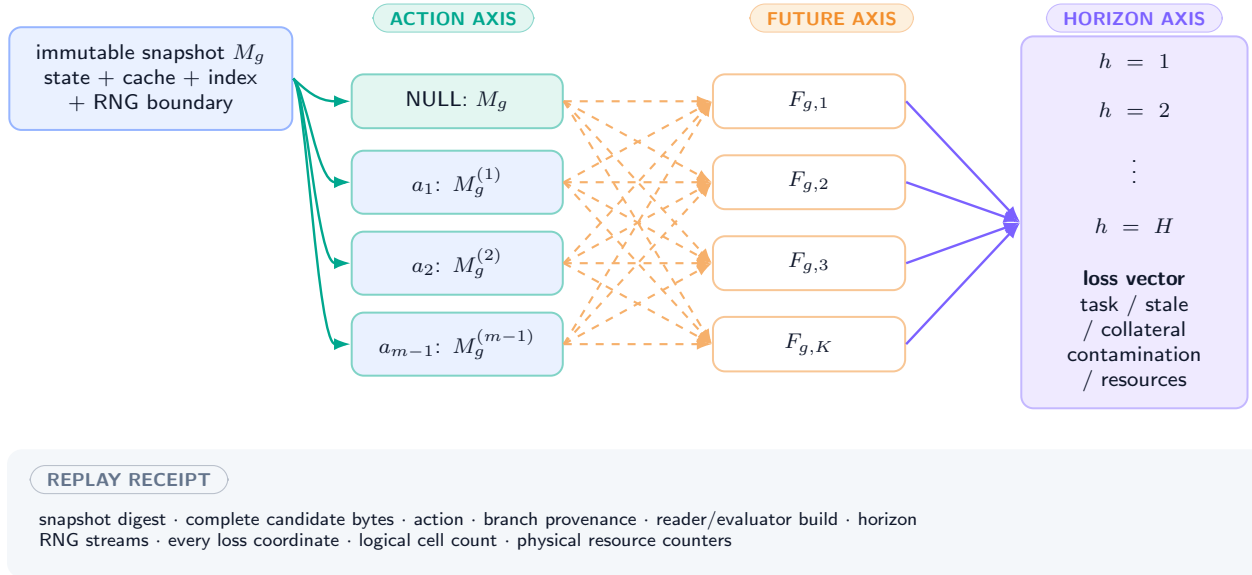
The complete replay array is

$$Y_g = (Y_{g,a,k,h} : a \in \mathcal{A}_g, 1 \leq k \leq K_g, 1 \leq h \leq H), \quad (17)$$

with action, future, and horizon axes. NULL occupies a full action slice. It is not a missing baseline.

Action-conditioned replay is a complete $|\mathcal{A}| \times K \times H$ outcome tensor

each action-future slice begins from the same immutable snapshot; NULL is a full identity-action slice



Batching and caching may reduce wall time only when exact reuse is proved. They do not remove a logical cell from the registered tensor.

Figure 4: **Action-conditioned replay tensor (DEFINITION / training only).** Every feasible write and NULL starts from one immutable snapshot, is evaluated on every grouped future, and contributes all registered horizon terms. The visible $|\mathcal{A}| \times K \times H$ lattice is logical work even when kernels batch it.

5.2 Loss vector and scalarization

For horizon step h , let

$$\ell_{g,a,k,h} = (\ell^{\text{task}}, \ell^{\text{stale}}, \ell^{\text{coll}}, \ell^{\text{cont}}, \ell^{\text{sec}}, \ell^{\text{lat}}, \ell^{\text{traffic}})_{g,a,k,h} \in \mathbb{R}_+^7. \quad (18)$$

The coordinates respectively measure task error, obsolete-value leakage, damage to unrelated information, admitted-content contamination, security/policy harm not already removed by hard infeasibility, latency, and memory/data traffic. The event-level loss is

$$L_g(a, F_{g,k}) = \sum_{h=1}^H \gamma^{h-1} \langle \lambda, \ell_{g,a,k,h} \rangle + \lambda_{\text{write}} \mathbf{1}[a \neq 0] + \lambda_{\text{null}} \mathbf{1}[a = 0] D_{g,k}^{\text{miss}} + \langle \lambda_{\text{res}}, \mathfrak{R}_{g,a,k} \rangle, \quad (19)$$

where $\gamma \in (0, 1]$, all weights are nonnegative and fixed before evaluation, D^{miss} prices a missed useful update without presuming every NULL is wrong, and $\mathfrak{R}$ is the replay resource vector. A hard safety violation should normally remove an action before scoring rather than be traded against task loss.

Term	Operational measurement	Boundary
Task loss	Frozen evaluator on future task/output	Evaluator error audited separately
Stale leakage	Old value emitted, selected, or remains authoritative when superseded	Must distinguish unknown from stale
Collateral damage	Positive loss change on unrelated facts/tasks versus the common snapshot	Neural cross-row effects measured, not assumed absent
Contamination	Severity of false, stale, malicious, or wrongly scoped authority	Requires truth/source contract
Write charge	Fixed cost for any committed row	Prevents free churn
NULL miss	Opportunity cost only when future evidence establishes a useful feasible write	Does not make abstention an error by default
Resources	Tokens, FLOPs, bytes, traffic, latency, energy, archive/index and validation work	Teacher and deployment ledgers stay separate

Table 3: Loss contract. Every coordinate and scalarization weight is preregistered; changing it changes the estimand.

DEFINITION

Definition 5.1 (Conditional horizon risk). For target continuation law $P_g(\cdot \mid \tilde{S}_g)$,

$$R_g(a) = \mathbb{E}_{F \sim P_g} \left[L_g(a, F) \mid \tilde{S}_g \right]. \quad (20)$$

The grouped empirical risk under branch law Q_g is

$$\hat{R}_g(a) = \sum_{k=1}^{K_g} w_{g,k} L_g(a, F_{g,k}). \quad (21)$$

The notation separates sampling and target distributions. When $Q_g = P_g$ and the group is exchangeable, $\hat{R}_g$ estimates the target risk. When $Q_g \neq P_g$, it estimates the synthetic or proposal-law risk unless corrected under support assumptions.

5.3 Paired action differences

Because all actions share each future branch, the paired difference

$$D_{g,k}(a, b) = L_g(a, F_{g,k}) - L_g(b, F_{g,k}) \quad (22)$$

directly estimates relative risk and cancels branch difficulty. Its variance for uniform independent branches is

$$\frac{1}{K_g} [\text{Var}(L_a) + \text{Var}(L_b) - 2 \text{Cov}(L_a, L_b)]. \quad (23)$$

Pairing reduces variance when action losses are nonnegatively correlated on the same future. It does not repair a biased branch law or invalid counterfactual replay.

5.4 Replay integrity

DESIGN PRESCRIPTION A valid replay record binds the source snapshot, complete candidate bytes, action, future branch, reader/evaluator build, horizon, RNG streams, loss coordinates, and resource increments. Action order is randomized or counterbalanced when shared hardware state could create interference. Each branch begins from an independently restored snapshot; caches and derived indexes are either restored or rebuilt under the selected action. A replay that omits NULL, prunes an action after seeing its outcome, or lets one branch inherit another branch’s state is invalid.

6 Expected-Risk Oracle, Current-Risk Comparator, and Qualification Gate

6.1 Three distinct decision objects

The empirical expected-risk oracle for group g is

$$\hat{a}_g^{\text{ER}} \in \arg \min_{a \in \mathcal{A}_g} \hat{R}_g(a). \quad (24)$$

It is an evaluation rule using training-only future branches. It is not deployed and it is not a learned policy.

The current-risk comparator is fixed independently:

$$a_g^{\text{cur}} \in \arg \min_{a \in \mathcal{A}_g} C_g^{\text{cur}}(a; \tilde{S}_g), \quad (25)$$

where C^{cur} may include current reconstruction, verification, immediate stale suppression, and write cost but no realized continuation. Its exact definition, tie rule, and parameters are preregistered. NULL is also reported independently, even if $a^{\text{cur}} = 0$.

Finally, the per-future hindsight action

$$a_{g,k}^{\text{hind}} \in \arg \min_a L_g(a, F_{g,k}) \quad (26)$$

is an unattainable information ceiling. Averaging its loss answers a different question from choosing one action that minimizes expected loss before the branch is known.

6.2 Expected oracle gap

Define group-level advantages over current risk and NULL:

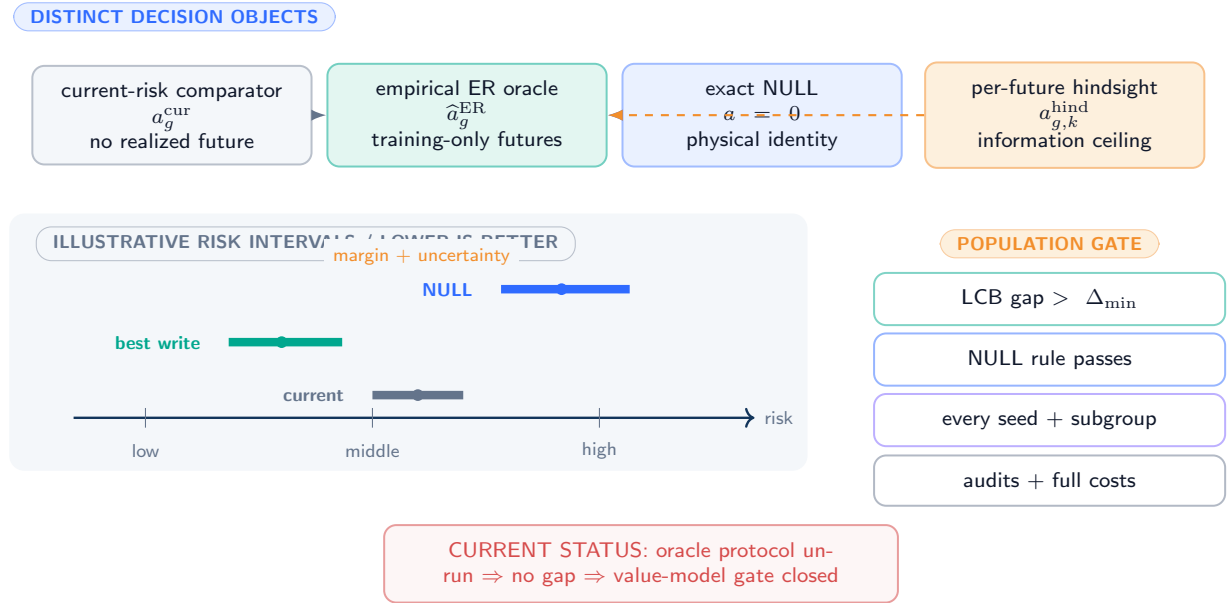
$$G_g^{\text{cur}} = R_g(a_g^{\text{cur}}) - \min_{a \in \mathcal{A}_g} R_g(a), \quad (27)$$

$$G_g^0 = R_g(0) - \min_{a \in \mathcal{A}_g} R_g(a). \quad (28)$$

For a population of fresh prefix groups $g \sim \mathcal{P}$, the target gaps are $G^{\text{cur}} = \mathbb{E}_g G_g^{\text{cur}}$ and $G^0 = \mathbb{E}_g G_g^0$. An oracle can have a positive advantage over current-risk writing but none over NULL, or vice versa. Both comparisons are required.

The expected-risk oracle gate requires two comparators and simultaneous uncertainty

illustrative geometry only — no risk, interval, gap, or passing seed is measured in this manuscript



A positive gap over current risk need not imply superiority to NULL. The registered relation to both baselines is reported separately.

Figure 5: **Oracle qualification geometry (PLANNED TEST)**. The empirical expected-risk oracle, current-risk comparator, and NULL are distinct. A stable simultaneous lower confidence bound must exceed the preregistered materiality margin on fresh groups and every required seed; otherwise the controller gate stays closed. No such gap is reported here.

6.3 Weighted concentration within one group

PROVED UNDER STATED ASSUMPTIONS

Theorem 6.1 (Simultaneous weighted Hoeffding bound). Assume 4.2, let $m_g = |\mathcal{A}_g|$, and suppose $0 \leq L_g(a, F) \leq L_{\max}$. Conditional on $\tilde{S}_g$, with probability at least $1 - \delta$,

$$\max_{a \in \mathcal{A}_g} \left| \hat{R}_g(a) - R_{Q_g}(a) \right| \leq L_{\max} \sqrt{\frac{\sum_{k=1}^{K_g} w_{g,k}^2}{2} \log \frac{2m_g}{\delta}} =: \varepsilon_g(\delta). \quad (29)$$

Proof. For fixed a , weighted Hoeffding gives $\mathbb{P}(|\sum_k w_k (L_{a,k} - \mathbb{E} L_{a,k})| > \epsilon) \leq 2 \exp[-2\epsilon^2 / (L_{\max}^2 \sum_k w_k^2)]$. A union bound over m_g actions makes the failure probability at most δ . Solving for ϵ gives Eq. 29. $\square$

The bound controls action selection and makes the multiple-action penalty explicit. It centers on R_{Q_g} , not R_{P_g} , when the branch law differs from deployment.

PROVED UNDER STATED ASSUMPTIONS

Corollary 6.2 (Stable empirical action). *Let a_g^* be the unique minimizer of R_{Q_g} and*

$$\Delta_g = \min_{a \neq a_g^*} (R_{Q_g}(a) - R_{Q_g}(a_g^*)). \quad (30)$$

If $\Delta_g > 2\varepsilon_g(\delta)$, then $\hat{a}_g^{\text{ER}} = a_g^$ with probability at least $1 - \delta$.*

Proof. On the simultaneous event, the optimal estimate can increase by at most ε_g and a competitor can decrease by at most ε_g . A true gap larger than $2\varepsilon_g$ cannot reverse. $\square$

Variance-sensitive empirical Bernstein intervals may be used under their stated conditions [7], but the estimator, finite-sample constants, action multiplicity, and calibration split must be frozen. Selecting whichever interval makes the gap significant is not valid.

6.4 Population gate

Protocol 6.3 (Oracle-gap qualification). **PLANNED TEST** For each fixed seed, estimate paired group-level advantages on a fresh namespace. Construct simultaneous confidence intervals for current-risk and NULL advantages across actions, primary subgroups, horizons, and required seeds using a preregistered group-level procedure. The expected-risk oracle qualifies only if:

1. the lower confidence bound for G^{cur} exceeds materiality margin $\Delta_{\min}^{\text{cur}} > 0$;
2. the registered relation to NULL passes its separate noninferiority or superiority rule;
3. action rankings, sign, stale leakage, and collateral damage are stable on every required seed and primary subgroup;
4. no leakage, support, evaluator, or replay audit fails; and
5. full teacher and deployment costs are published.

If any condition fails, Q_ψ is not trained.

UNVALIDATED Protocol 6.3 has not been executed. This manuscript defines the gate and proves conditional properties; it does not claim an oracle advantage of any magnitude.

7 Uncertainty-Aware Conservative Admission

7.1 Simultaneous intervals

For each action, let $[L_g(a), U_g(a)]$ be an interval for $R_g(a)$ constructed without final-set tuning. The required coverage is simultaneous:

$$\mathbb{P}(R_g(a) \in [L_g(a), U_g(a)] \text{ for every } a \in \mathcal{A}_g) \geq 1 - \delta. \quad (31)$$

Marginal 95% intervals for many actions do not imply 95% coverage of the selected action. Bonferroni, finite-class concentration, or another frozen simultaneous procedure must charge action selection. If thresholds are also chosen on calibration data, final coverage is evaluated on untouched groups.

Let the conservative write candidate be

$$i_g^+ \in \arg \min_{a \in \mathcal{A}_g \setminus \{0\}} U_g(a), \quad (32)$$

with a fixed tie rule. For required benefit margin $\mu \geq 0$, define

$$\pi_{\text{cert}}(\tilde{S}_g) = \begin{cases} i_g^+, & U_g(i_g^+) + \mu < L_g(0), \\ 0, & \text{otherwise.} \end{cases} \quad (33)$$

If there is no feasible write, the rule returns NULL.

Conservative admission compares simultaneous intervals, not raw scores

commit a write only when its upper risk bound plus a frozen margin is below the lower bound for exact NULL

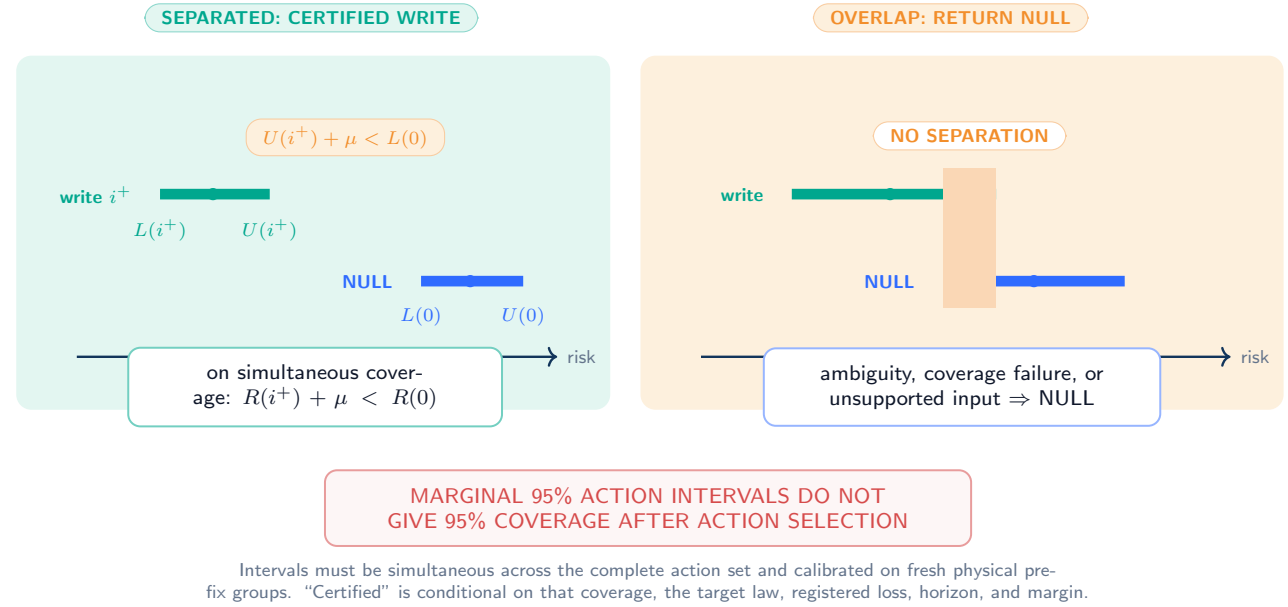


Figure 6: **Conservative UCB/LCB admission (PROVED UNDER STATED ASSUMPTIONS under simultaneous coverage)**. The best feasible write commits only when its upper risk bound plus margin is below the lower bound for NULL. Overlap, multiplicity, or failed calibration returns NULL.

PROVED UNDER STATED ASSUMPTIONS

Theorem 7.1 (Certified improvement over NULL). *Under Eq. 31, every non-NULL action selected by Eq. 33 satisfies*

$$R_g(i_g^+) + \mu < R_g(0) \quad (34)$$

with probability at least $1 - \delta$.

Proof. On the simultaneous-coverage event, $R_g(i_g^+) \leq U_g(i_g^+)$ and $L_g(0) \leq R_g(0)$. The commit rule therefore yields $R_g(i_g^+) + \mu \leq U_g(i_g^+) + \mu < L_g(0) \leq R_g(0)$. The event has probability at least $1 - \delta$. $\square$

This result certifies improvement over NULL under the interval assumptions. It does not prove that i_g^+ is the globally best write. If target correctness matters, require the additional separation

$$U_g(i_g^+) + \eta < \min_{a \in \mathcal{A}_g \setminus \{0, i_g^+\}} L_g(a) \quad (35)$$

for a prespecified $\eta \geq 0$, or report the action as beneficial but target-ambiguous.

7.2 NULL calibration

NULL is both a physical identity action and a statistical reject option, related to classical error-reject tradeoffs [1]. It needs its own calibration. A model can achieve low write error by returning NULL everywhere, or high recall by writing everywhere. Required outputs include:

- empirical interval coverage for NULL and non-NULL actions separately;
- precision and recall of NULL on preregistered write-worthy and null-worthy groups;
- risk-coverage curves as μ varies only on calibration data;
- subgroup calibration by horizon, capacity, action count, source reliability, and event type; and
- an ambiguity rate for groups where intervals do not separate.

DEFINITION

Definition 7.2 (Write-worthy, null-worthy, and ambiguous). For materiality margin μ and true best-write risk $R_g^W = \min_{a \neq 0} R_g(a)$, a group is write-worthy if $R_g^W + \mu < R_g(0)$, null-worthy if $R_g(0) + \mu < R_g^W$, and ambiguous otherwise.

Ambiguous groups are not forced into a favorable binary label. A false write is a non-NULL decision on a null-worthy group; a false NULL is NULL on a write-worthy group; a wrong-target write has regret above a fixed target tolerance even when some write is beneficial.

7.3 Uncertainty sources

Source	Required treatment	Failure boundary
Finite future branches	Concentration or group bootstrap with action multiplicity	Small K_{eff} makes intervals wide
Finite prefix groups	Group-level resampling; no branch-level pseudoreplication	Shared prefixes are not independent units
Value-model uncertainty	Calibrated ensemble, quantile, or conformal-style method fixed before final use	Neural scores are not confidence bounds by naming
Distribution shift	Shift audit or declared robust set	Nominal coverage need not transfer
Evaluator uncertainty	Error/sensitivity analysis in the loss scale	Miscalibration can certify the wrong action
Constructor randomness	Freeze seeds or integrate over recorded draws	Selection can exploit lucky candidates

Table 4: Uncertainty sources. Aleatoric branch variation, estimation error, model error, and evaluator error are not interchangeable.

DESIGN PRESCRIPTION Any monitoring failure, coverage shortfall, unsupported action, or out-of-scope group switches the deployed selector to NULL-only. This is a design rule, not evidence that a future implementation detects every shift.

8 Identification, Distribution Shift, and Impossibility

8.1 What full grouped replay identifies

Let $Y_g(a, F)$ be the complete potential replay outcome for action a and continuation F . Under Assumptions 4.2–4.4, full action replay from one common snapshot identifies $R_{Q_g}(a) = \mathbb{E}_{Q_g} L(Y_g(a, F))$ for every feasible action, up to sampling error. If $Q_g = P_g$, this is the deployment conditional risk. If the future is action dependent, identification instead requires valid samples from each interventional law $P_g(F \mid \text{do}(A = a), \tilde{S}_g)$.

The conclusion is narrow. Grouped futures do not identify a different deployment branch law, an action absent from support, an evaluator’s latent truth, or the value of features that would be unavailable before the action.

8.2 Missing support

PROVED UNDER STATED ASSUMPTIONS

Theorem 8.1 (Unsupported-action non-identification). *Suppose action $a^\dagger$ is never replayed and has zero logging probability on a set of prefix states with positive target probability. Without an additional outcome model restriction, $R(a^\dagger)$ is not identified from the observed data.*

Proof. Construct two data-generating laws that agree on prefix states, logging actions, and every observed action outcome. Under the first law set the unobserved loss of $a^\dagger$ to 0 on the unsupported set; under the second set it to $L_{\max}$. Both induce exactly the same observed distribution because $a^\dagger$ is never taken there, yet their target risks differ by $L_{\max}$ times the set’s positive probability. Therefore the risk is not identified. $\square$

This result is why a learned shortlist cannot silently replace full-action replay in the first oracle test. Shortlist recall must itself be qualified before omitted actions can be ignored.

8.3 Branch-distribution shift

PROVED UNDER STATED ASSUMPTIONS

Theorem 8.2 (Total-variation risk and gap sensitivity). *Let $0 \leq L(a, F) \leq L_{\max}$ and define $d = \text{TV}(P, Q)$. Then for every action*

$$|R_P(a) - R_Q(a)| \leq L_{\max} d. \quad (36)$$

For any two actions a, b ,

$$|\{R_P(a) - R_P(b)\} - \{R_Q(a) - R_Q(b)\}| \leq 2L_{\max} d. \quad (37)$$

Proof. For a measurable function in $[0, L_{\max}]$, the difference of expectations under P and Q is at most $L_{\max} \text{TV}(P, Q)$. Apply this once for Eq. 36 and twice with the triangle inequality for Eq. 37. $\square$

A synthetic oracle gap smaller than $2L_{\max} d$ can reverse under a deployment law within that distance. More synthetic branches cannot remove this bias. Distributionally robust objectives can optimize over a declared ambiguity set [2]; they do not guarantee coverage of an undeclared shift.

8.4 Evaluator misspecification

Let L^* be the target loss and $\tilde{L}$ the evaluator loss.

PROVED UNDER STATED ASSUMPTIONS

Proposition 8.3 (Evaluator-error bound). *If $|\tilde{L}(a, F) - L^*(a, F)| \leq \epsilon_{\text{eval}}$ almost surely for every action, then*

$$|\tilde{R}(a) - R^*(a)| \leq \epsilon_{\text{eval}}, \quad (38)$$

and every pairwise action gap can differ by at most $2\epsilon_{\text{eval}}$.

Proof. Take expectations of the pointwise bound; apply the triangle inequality to two actions. $\square$

Without any link between evaluator and target loss, even the sign of an action advantage is not identified: two latent target-loss laws can agree on every evaluator output and reverse which action is better.

8.5 Leakage and confounding

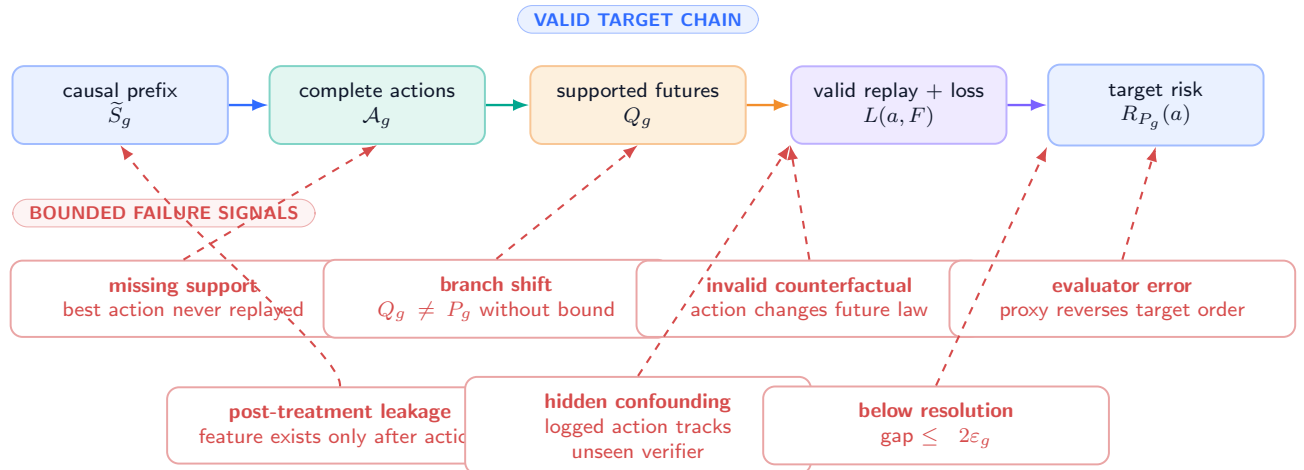
Counterexample 8.4 (Post-treatment feature). The write changes a future hidden state. A feature extractor then reads that hidden state and supplies it to the value model as if it were a pre-action feature. The feature perfectly predicts which action produced the favorable replay, but it is unavailable before selection. Offline accuracy can be one while the deployed input is undefined. This violates $\mathcal{D}_{n,j}$ -measurability.

Counterexample 8.5 (Hidden confounding in logged actions). A production heuristic writes only when an unrecorded verifier is confident. Written cases later have lower loss than NULL cases. The association can reflect verifier confidence rather than the write. Without randomization, observed confounder adjustment, or valid instrumental structure, the action effect is not identified.

Counterexample 8.6 (Near-zero action gap). Two actions have risks differing by $\Delta \leq 2\epsilon_g(\delta)$. Either action can be the empirical minimizer on the simultaneous concentration event. Treating the selected index as a reliable class label creates false precision; the group should be ambiguous or receive a soft value target with uncertainty.

Identification failures are data and causal defects, not capacity problems

each failure breaks a different link between grouped replay, target risk, and deployable value



More branches reduce variance only around the law actually sampled. A larger neural controller cannot manufacture an unsupported outcome, remove leakage, validate a counterfactual, or link a proxy evaluator to truth.

Figure 7: **Identification and failure counterexamples.** Coral marks bounded failure signals: missing action support, branch shift, post-treatment leakage, hidden confounding, evaluator error, and action gaps below sampling resolution. More controller capacity does not repair these defects.

8.6 Identification ledger

Target	Identified when	Not identified when
$R_{Q_g}(a)$	Full valid replay; exchangeable branches; fixed loss	Action absent, replay invalid, dependence ignored
$R_{P_g}(a)$	$Q_g = P_g$ or valid supported correction	Synthetic/model law differs without a bound
Action effect	Exogenous common future or interventional simulator	Action changes future and one observed branch is reused
Future-blind predictability	Fresh prefix groups; causal features only	Future-derived or post-treatment feature enters model
Truth-linked risk	Evaluator link audited	Proxy can reverse target ordering
Long-run value	Repeated-write state distribution evaluated	One-step groups only

Table 5: Identification summary. Each target has a separate premise bundle.

9 Future-Blind Value Estimation and Selection

9.1 A value model is not an oracle and not a policy

The expected-risk oracle in Eq. 24 sees training-only future branches. A deployable value model may see only the causal information available before action selection. Let

$$X_{g,a} = \phi(\tilde{S}_g, a, c_{g,a}) \quad (39)$$

be a typed, versioned feature record with $c_{g,0} = \emptyset$. Every coordinate of $X_{g,a}$ must be measurable with respect to $\mathcal{D}_g$ from Eq. 7. Define the conditional value target

$$q^*(x) = \mathbb{E}[L_g(a, F) \mid X_{g,a} = x]. \quad (40)$$

A model $Q_\psi(X_{g,a})$ approximates Eq. 40; it does not receive F , a replay outcome, an oracle label computed from the same group, or any post-action state.

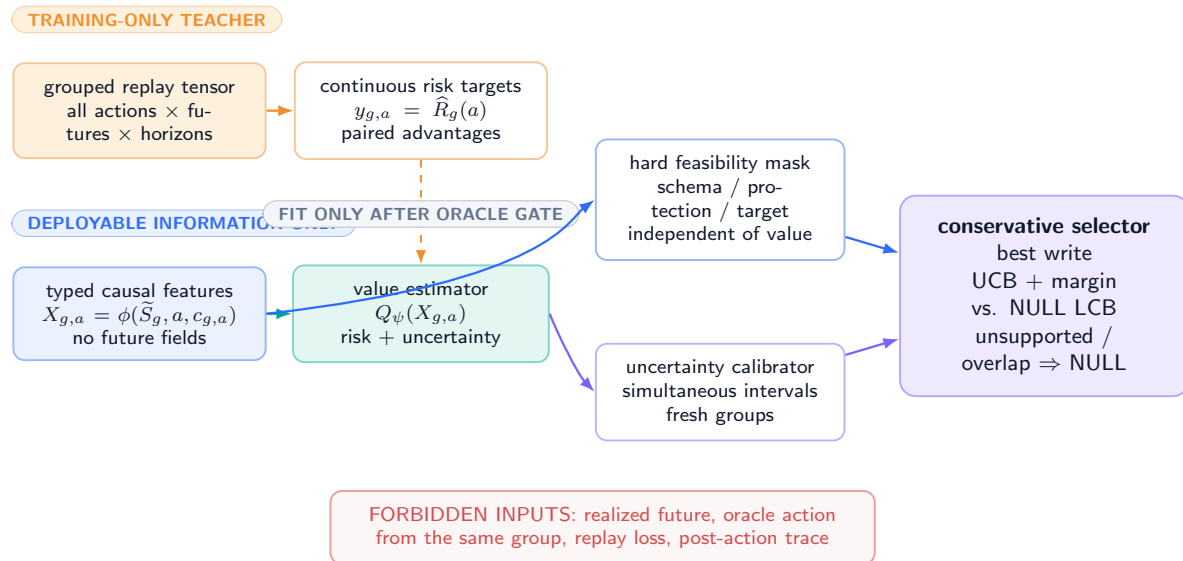
Three objects must remain separate:

1. the *value estimator* Q_ψ , which predicts action risk;
2. the *feasibility mask* $\text{Valid}(\tilde{S}_g, a, c_{g,a})$, which enforces the row contract and hard constraints; and
3. the *selector*, which compares calibrated action intervals and may return NULL.

Calling the first object a policy erases two independently testable interfaces. This manuscript specifies all three but instantiates none.

Future-blind value estimation keeps teacher, model, mask, and selector separate

the architecture is a planned protocol; no component below is trained or empirically calibrated here



Low regression error is not a policy result. Report complete-set regret, NULL behavior, interval coverage, subgroup stability, negative controls, and cost separately.

Figure 8: **Future-blind value architecture (PLANNED TEST)**. Training-only grouped futures create action-risk targets. Deployment exposes only typed prefix state, action identity, and candidate metadata. The learned value, feasibility mask, uncertainty calibrator, and conservative selector remain separately auditable. No value model or policy is trained here.

9.2 Training targets and losses

The primary target is the continuous paired risk vector or its frozen scalarization, not merely the empirical argmin. For a training group, define $y_{g,a} = \widehat{R}_g(a)$ and paired advantages $d_{g,a} = y_{g,a} - y_{g,0}$. A preregistered value objective may combine

$$\begin{aligned} \mathcal{L}_{\text{val}}(\psi) = & \sum_{g,a} \omega_{g,a} \rho_{\kappa}(Q_{\psi}(X_{g,a}) - y_{g,a}) \\ & + \lambda_{\text{rank}} \sum_g \sum_{a < b} \log(1 + \exp\{-s_{g,a,b}[Q_{\psi}(X_{g,b}) - Q_{\psi}(X_{g,a})]\}), \end{aligned} \quad (41)$$

where ρ_{κ} is the Huber loss, $s_{g,a,b} \in \{-1, 0, 1\}$ is a margin-aware ordering label, and ambiguous pairs receive $s = 0$ and no ranking charge. Group weights are fixed without using final outcomes. A heteroscedastic, quantile, ensemble, or conformal-style head may estimate uncertainty, but its coverage is an empirical property, not a consequence of architecture naming.

Regression error, action ranking, interval calibration, and selector regret are reported separately. Low mean squared error can coexist with wrong top actions near a decision boundary. Conversely, correct ranking can coexist with badly calibrated risk magnitudes and unsafe commit thresholds.

9.3 Value error and decision regret

For a fixed prefix group, let $a_g^* \in \arg \min_{a \in \mathcal{A}_g} R_g(a)$ and let

$$\widehat{a}_g \in \arg \min_{a \in \mathcal{A}_g} \widehat{Q}_g(a), \quad \widehat{Q}_g(a) = Q_{\psi}(X_{g,a}), \quad (42)$$

with a fixed tie rule.

PROVED UNDER STATED ASSUMPTIONS

Theorem 9.1 (Uniform value error implies bounded action regret). *If $|\widehat{Q}_g(a) - R_g(a)| \leq \epsilon_g$ for every $a \in \mathcal{A}_g$, then*

$$0 \leq R_g(\widehat{a}_g) - R_g(a_g^*) \leq 2\epsilon_g. \quad (43)$$

If the true best action is unique and separated from every competitor by more than $2\epsilon_g$, then $\widehat{a}_g = a_g^$.*

Proof. Optimality of $\widehat{a}_g$ for the estimated values gives $\widehat{Q}_g(\widehat{a}_g) \leq \widehat{Q}_g(a_g^*)$. Hence

$$R_g(\widehat{a}_g) \leq \widehat{Q}_g(\widehat{a}_g) + \epsilon_g \leq \widehat{Q}_g(a_g^*) + \epsilon_g \leq R_g(a_g^*) + 2\epsilon_g.$$

The separation statement follows because choosing a competitor would contradict this bound. $\square$

The premise is simultaneous and prefix-conditional. An average validation error does not establish it. Under branch-law shift and evaluator error, a useful sensitivity statement combines Theorem 8.2 and Proposition 8.3: a model error bound ϵ_g , distribution distance d_g , and evaluator error ϵ_{eval} can permit pairwise ordering error on the scale

$$2\epsilon_g + 2L_{\max}d_g + 2\epsilon_{\text{eval}}. \quad (44)$$

This is a diagnostic budget, not a claim that any term is currently bounded.

9.4 Training gate and abstaining deployment

PLANNED TEST Value-model training may begin only after Protocol 6.3 qualifies a reproducible, material expected-risk advantage on fresh prefix groups. The data split is by physical prefix group, source namespace, and registered temporal boundary. Hyperparameters, feature schemas, action ordering, calibration method, and commit margin are frozen before final evaluation.

At deployment, unsupported schemas, invalid candidates, failed monitoring, or nonseparating intervals return NULL. The selection rule is Eq. 33, not an unconstrained argmin of a neural score. A later learned shortlist would require an independent recall guarantee against the complete feasible action set before it could reduce replay or scoring work.

UNVALIDATED There is no qualified oracle gap, trained Q_{ψ} , learned shortlist, calibrated interval model, or deployed learned selector in the frozen evidence record. The entire section is conditional theory and an unopened experimental protocol.

10 Metrics, Splits, Leakage Audits, and Reproducibility

10.1 Decision metrics

Let a_g^* minimize the registered target risk and let π_g be the evaluated selector. The primary per-group action regret is

$$\text{Reg}_g(\pi) = R_g(\pi_g) - R_g(a_g^*) \geq 0. \quad (45)$$

When a fixed scale $s_g > 0$ is preregistered, normalized regret is Reg_g/s_g ; a scale chosen after seeing results is prohibited. Report the mean, median, upper quantiles, worst registered subgroup, and a group-level confidence interval. Also report:

- top-1 action agreement and top- k ranking metrics, with ambiguous groups separated;
- excess risk relative to current risk and relative to NULL;
- write precision, write recall, NULL precision, and NULL recall using the margin-aware classes in Section 7;
- wrong-target rate conditional on writing;
- stale leakage, collateral damage, contamination, and resource coordinates before scalarization;
- interval coverage, width, risk-coverage curves, calibration error, and ambiguity rate; and
- stability across seeds, horizons, capacities, event types, source reliability, and action-set size.

Future loss is always computed by action-conditioned replay. Current-state reconstruction may be a diagnostic but cannot substitute for Eq. 20. Accuracy on oracle action labels is secondary because labels are unstable when risk differences lie inside uncertainty intervals.

10.2 Split hierarchy

DESIGN PRESCRIPTION A physical causal prefix, all its candidate actions, and all its future branches form one indivisible group. The primary split hierarchy is:

1. source namespace and temporal boundary;
2. physical prefix group;
3. registered scenario family or generator seed; and
4. only then, individual action and branch records within the assigned group.

No branch, paraphrase, candidate, cached trace, or derived oracle label from one physical prefix may cross train, calibration, and final partitions. The final partition is read once by the frozen evaluation entry point.

10.3 Leakage and falsification audits

Audit	Receipt	Passing condition	Failure action
Prefix provenance	Source, byte hash, cutoff, event index	Feature timestamps precede action	Invalidate group
Group isolation	Physical-prefix identifier closure	No identifier or derivative crosses split	Rebuild split
Future blindness	Static graph plus runtime access log	No branch/outcome/post-action field reaches model	Reject model
Replay restoration	Snapshot, cache, index, RNG receipts	Every action-branch begins identically	Recompute tensor
Action completeness	Deterministic feasible-set manifest	Every valid action and NULL priced	Reject oracle test
Evaluator freeze	Build hash and blinded final entry	No final-set tuning or hidden revision	Restart final evaluation
Label permutation	Group-preserving shuffled future labels	Performance collapses to registered null behavior	Investigate leakage
Prefix destruction	Mask registered causal signal only	Change agrees with preregistered sensitivity	Mark model suspect
Seed replication	Independent generator and constructor seeds	Sign and primary subgroup gates pass every seed	Stop progression
Cost reconciliation	Logical counts plus measured counters	Ledger totals match trace within tolerance	Reject efficiency claim

Table 6: Minimum audit suite. Receipts are immutable artifacts, not narrative assurances.

The shuffled-label audit preserves group structure; branch-level shuffling that destroys obvious correlations can be too easy. A stronger negative control swaps future sets only among matched prefixes without changing action metadata. Success above the registered null range indicates leakage, unmodeled group identity, or a flawed split.

10.4 Calibration protocol

For interval level $1 - \alpha$, empirical coverage on a final set $\mathcal{G}_{\text{test}}$ is

$$\widehat{\text{Cov}}_{1-\alpha} = \frac{1}{|\mathcal{G}_{\text{test}}|} \sum_{g \in \mathcal{G}_{\text{test}}} \mathbf{1}[R_g(a) \in [L_g(a), U_g(a)] \ \forall a \in \mathcal{A}_g]. \quad (46)$$

This group-level simultaneous coverage, not marginal action coverage, is the primary quantity. If $R_g(a)$ itself is estimated, its teacher interval and the model interval are reported separately; treating a noisy teacher estimate as exact understates uncertainty.

For a binary write-worthiness probability p_g and label z_g , expected calibration error is reported only with bins fixed on calibration data, alongside proper scores and reliability diagrams. Because binning can conceal local failure, primary commit safety is judged by realized risk and simultaneous interval coverage.

10.5 Reproducibility record

Every run record binds code and environment identifiers, immutable data manifests, feature schema, group split, action manifest, constructor and evaluator versions, future-generation provenance, all RNG seeds, scalarization weights, interval method, cost counters, and output hashes. Failed and stopped runs remain indexed. Aggregates are regenerated from row-level records; no number is transcribed manually into a paper table.

PLANNED TEST These metrics and audits define future acceptance criteria. None has been run for a predictive admission value model in the evidence available to this manuscript.

11 Complete Teacher and Deployment Cost Ledger

11.1 Logical work cannot be hidden by batching

For group g , write $m_g = |\mathcal{A}_g|$, with one NULL action, K_g future branches, and horizon H_g . A logical replay cell is one tuple (g, a, k, h) whose action-conditioned state produces the registered horizon outputs and loss coordinates.

PROVED UNDER STATED ASSUMPTIONS

Proposition 11.1 (Exact logical replay count). *Under full action support and complete grouped replay, the number of horizon cells is*

$$N_{\text{cell}} = \sum_{g \in \mathcal{G}} m_g K_g H_g. \quad (47)$$

The number of non-NULL complete candidate constructions is

$$N_{\text{cand}} = \sum_{g \in \mathcal{G}} (m_g - 1). \quad (48)$$

If event-recursive teaching branches at every event, the corresponding tree count replaces m_g by the number of supported prefix-action paths and can grow multiplicatively.

Proof. For fixed g , the Cartesian product $\mathcal{A}_g \times \{1, \dots, K_g\} \times \{1, \dots, H_g\}$ has $m_g K_g H_g$ elements. Summing disjoint groups proves Eq. 47. Exactly one action is NULL, so $m_g - 1$ complete non-NULL candidates are constructed. Event recursion evaluates supported paths rather than isolated labels; multiplying per-event branching factors gives the path count. $\square$

Vectorization, shared kernels, prefix caching, or parallel hardware may reduce wall time but not the logical outcome array required by the estimand. Any reuse is reported together with the assumptions that make it exact.

Complete cost ledger: logical teacher coverage and physical deployment work

tokens or accelerator time alone are incomplete; authoritative bytes, traffic, archives, indexes, validation, and retries remain visible

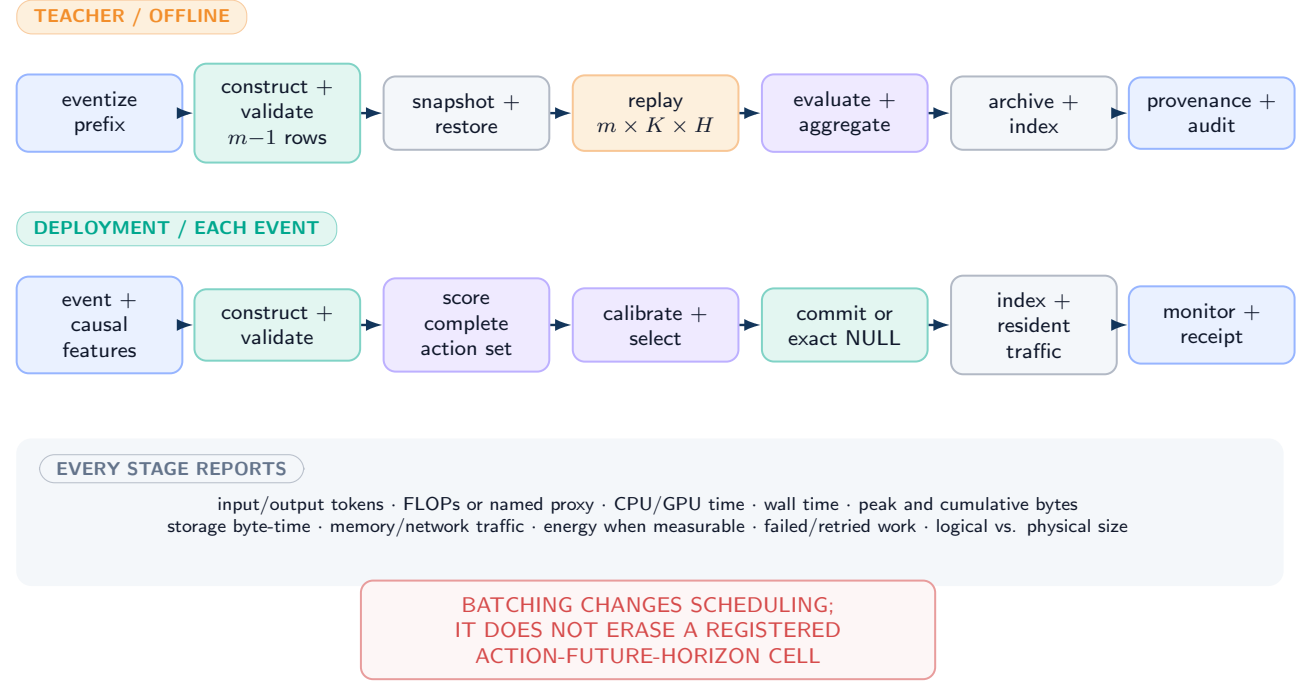


Figure 9: **End-to-end cost ledger (DEFINITION)**. The teacher pays eventization, every complete candidate, validation, $|\mathcal{A}| \times K \times H$ replay, evaluation, archives, indexes, and provenance. Deployment pays causal feature extraction, candidate construction, scoring, calibration, commit, and resident-state traffic. Batching changes scheduling, not logical coverage.

11.2 Teacher ledger

For each group, report separately:

$$\begin{aligned}
 C_g^{\text{teach}} = & C_g^{\text{event}} + \sum_{a \neq 0} \left(C_{g,a}^{\text{construct}} + C_{g,a}^{\text{validate}} \right) + C_g^{\text{snapshot}} \\
 & + \sum_{a \in \mathcal{A}_g} \sum_{k=1}^{K_g} \left(C_{g,a,k}^{\text{restore}} + C_{g,a,k}^{\text{replay}} + C_{g,a,k}^{\text{eval}} \right) \\
 & + C_g^{\text{aggregate}} + C_g^{\text{archive}} + C_g^{\text{index}} + C_g^{\text{provenance}}.
 \end{aligned} \tag{49}$$

The units ledger contains input/output tokens, model FLOPs or an explicitly named proxy, accelerator and CPU time, peak and cumulative bytes, storage byte-time, network and memory traffic, energy when measurable, wall time, and failed/retried work. Trusted assembly, parsing, validation, and evaluator calls are not free because they are outside the main model.

Teacher storage includes source snapshots, complete candidate bytes, future branches, action-conditioned traces or sufficient replay receipts, per-horizon loss vectors, resource counters, and provenance. Deduplication savings are reported with physical and logical sizes. Archive and index maintenance are charged to the design that needs them.

11.3 Deployment ledger

Deployment cost for one event is

$$\begin{aligned}
 C_g^{\text{deploy}} = & C_g^{\text{event}} + C_g^{\text{feature}} + \sum_{a \neq 0} \left(C_{g,a}^{\text{construct}} + C_{g,a}^{\text{validate}} \right) \\
 & + \sum_{a \in \mathcal{A}_g} C_{g,a}^{\text{score}} + C_g^{\text{calibrate}} + C_g^{\text{select}} + C_g^{\text{commit}} + C_g^{\text{index}} + C_g^{\text{monitor}}.
 \end{aligned} \tag{50}$$

If candidates are constructed only after a learned shortlist, the shortlist’s complete-set recall and the cost of generating the features it needs must be counted. Otherwise apparent savings can be caused by silently changing the feasible action set.

11.4 Fair comparisons

Compare systems at equal resident authoritative byte-time, equal archive availability, equal retrieval context, equal evaluator quality, and equal task horizon. Report both teacher-amortized and steady-state deployment views. A method that moves rows from authoritative memory into an uncharged archive, increases retrieval tokens, or relies on expensive future generation has not demonstrated a system-level efficiency gain.

For repeated deployment over T events, report

$$C_{1:T}^{\text{total}} = C^{\text{train}} + C^{\text{calibration}} + \sum_{t=1}^T C_t^{\text{deploy}} + C_{1:T}^{\text{resident}} + C_{1:T}^{\text{archive}}, \quad (51)$$

together with the chosen amortization horizon. Savings that change sign outside one chosen T are shown as a curve.

UNVALIDATED No teacher run, deployment run, or predictive-admission efficiency measurement is reported here. Equations 47–51 are accounting requirements for later work.

12 Counterexamples, Long-Run Pollution, and Stop Rules

12.1 Why an apparent oracle is insufficient

Counterexample 12.1 (Hindsight without conditional predictability). There are two writes a_1, a_2 and two equiprobable futures F_1, F_2 . Loss is zero when indices match and one otherwise. A per-future hindsight selector has zero loss, but both fixed actions have expected loss $1/2$, and no causal prefix feature predicts the future. The expected-risk oracle is tied and no learned admission rule can recover the hindsight gain. Thus an impressive hindsight ceiling is not an expected-risk oracle gap.

Counterexample 12.2 (Current-risk reversal). Action a_1 perfectly reconstructs the completed segment and has current cost zero; action a_2 has current cost 0.1. With probability 0.9, later tasks need the fact stored by a_2 and never query a_1 ’s fact. A frozen horizon loss can prefer a_2 even though the current-risk comparator prefers a_1 . Reversing the probabilities reverses the expected-risk order without changing current reconstruction. Current-risk quality therefore neither proves nor refutes future value.

Counterexample 12.3 (Short-horizon benefit, long-run pollution). A write reduces next-step task loss by 0.2 but overwrites a protected relation that is queried once in the next hundred steps, causing loss 1. An $H = 1$ evaluator selects the write; an $H = 100$ evaluator rejects it. Horizon is part of the estimand and cannot be chosen after inspecting this reversal.

Counterexample 12.4 (Invalid synthetic branch). A generator produces diverse continuations conditional on a textual prefix but omits an external process that makes a future depend on the chosen write. Replaying every action on those shared branches appears well supported. The counterfactual is nevertheless invalid because Assumption 4.4 fails; more generated branches concentrate around the wrong law.

Counterexample 12.5 (Hidden shortlist failure). A cheap heuristic proposes two of ten feasible targets and the teacher prices only those plus NULL. The omitted target is best on a rare but material subgroup. The learned controller matches its truncated teacher perfectly while incurring large complete-set regret. Unless shortlist recall was independently qualified, the result is not an oracle for $\mathcal{A}_g$.

12.2 Accumulation over repeated writes

Let $Z_t = 1$ indicate that the t th committed write introduces a registered contamination event that remains consequential within the lifecycle horizon. Let $\mathcal{H}_{t-1}$ be the pre-write history and suppose the monitoring argument establishes

$$\mathbb{P}(Z_t = 1 \mid \mathcal{H}_{t-1}, A_t \neq 0) \leq p_t. \quad (52)$$

PROVED UNDER STATED ASSUMPTIONS

Proposition 12.6 (Repeated-write pollution bound). *For any possibly adaptive sequence of at most T committed writes satisfying Eq. 52,*

$$\mathbb{P}\left(\bigcup_{t=1}^T \{Z_t = 1\}\right) \leq \sum_{t=1}^T p_t, \quad \mathbb{E}\left[\sum_{t=1}^T Z_t\right] \leq \sum_{t=1}^T p_t. \quad (53)$$

No independence assumption is required.

Proof. The union bound gives the first inequality. For the second, the tower property yields $\mathbb{E}Z_t = \mathbb{E}[\mathbb{E}(Z_t \mid \mathcal{H}_{t-1}, A_t \neq 0)] \leq p_t$ for committed writes; linearity of expectation completes the proof. $\square$

For constant $p_t = p$, keeping the union-bound budget below α requires $p \leq \alpha/T$. The bound can be loose, but it exposes a key scaling fact: a seemingly small per-write error is not automatically a small lifecycle risk. Dependencies, persistence duration, repair, and severity require richer stateful evaluation. This proposition does not establish any empirical p_t .

12.3 Mandatory stop rules

Observed condition	Scientific conclusion	Required action
Inherited semantic row gate failed or unqualified	Admission action has no accepted semantic substrate	Do not run oracle/controller claim
No reproducible material gap over current risk	Predictive selection has not earned a target	Stop before value-model training
No registered relation to NULL	Write benefit is not established	Keep NULL-only deployment
Gap changes sign by seed, subgroup, or horizon	Oracle effect is unstable or underspecified	Diagnose; no pooled promotion
Oracle gap exists but causal features do not rank actions	Future value is not predictably available	Stop controller progression
Intervals fail simultaneous coverage	Commit certificate is invalid	Return NULL; recalibrate on nonfinal data
Leakage, support, replay, or evaluator audit fails	Estimand is not identified by the run	Invalidate affected result
Long-run pollution exceeds budget	Short-horizon gain is insufficient	Extend horizon or reject controller
Full lifecycle cost erases registered benefit	System-level efficiency is unsupported	Report tradeoff; no efficiency claim
Only a learned shortlist is evaluated	Complete action regret is unknown	Restore full set or qualify recall first

Table 7: Stop rules. A stop is an informative outcome, not a request to weaken the gate.

PLANNED TEST The next admissible experiment, only after an independent semantic-interface qualification, is a single-event, fixed-payload, complete-action grouped replay with exact NULL, all candidates constructed before futures are opened, and every cost coordinate logged. If its expected-risk gap does not qualify, the experimental ladder ends. This manuscript neither executes nor authorizes that experiment.

13 Qualification Ladder and Theorem-to-Implementation Obligations

Conditional proofs constrain an implementation only when their premises are discharged by inspectable receipts. Figure 10 and Table 8 make those dependencies explicit. A later stage cannot retroactively validate an earlier one.

Serialized qualification ladder: every empirical gate may terminate the line

conditional proofs point to required receipts; they do not establish an implementation premise or authorize the next stage

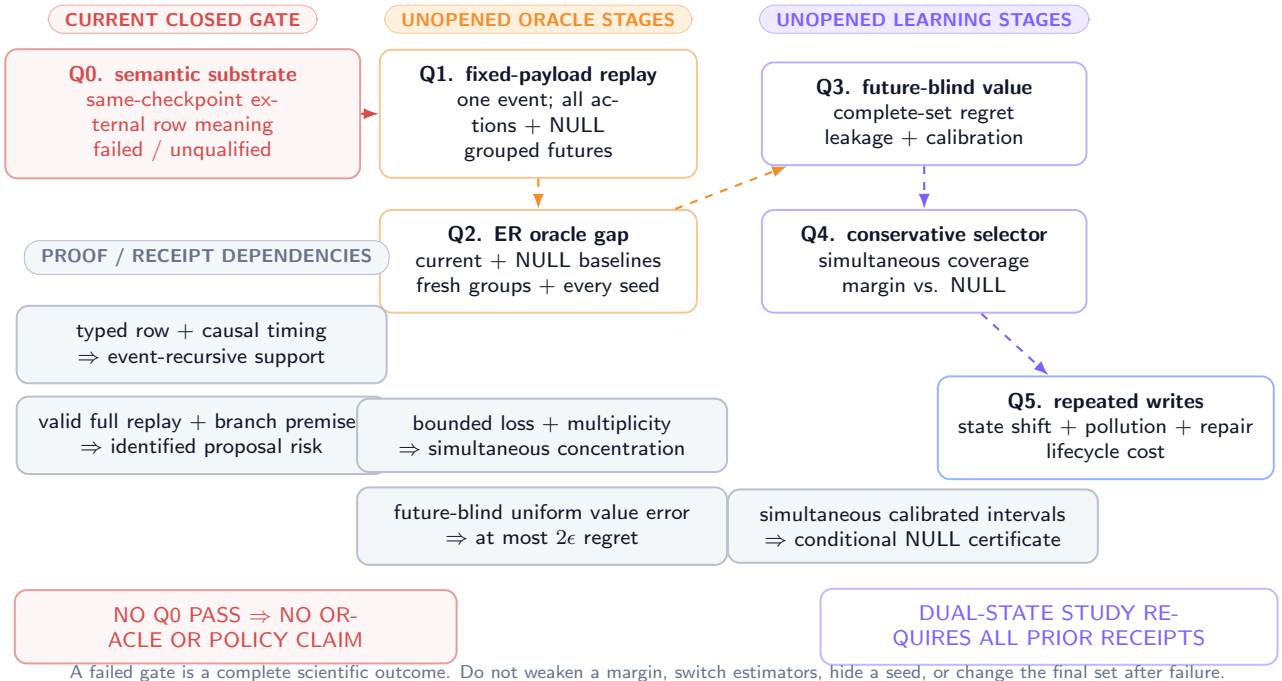


Figure 10: **Qualification ladder and proof dependencies.** The semantic-interface stop precedes fixed-payload oracle replay. Only a stable expected-risk gap permits future-blind value training; only calibrated low-regret selection permits repeated-write study. Coral gates are currently closed or unvalidated.

13.1 Proof dependency graph

The principal logical dependencies are:

typed row contract + causal timing $\implies$ event-recursive support,
 common prefix + valid full replay + branch assumptions $\implies$ identified empirical action risks,
 bounded loss + independent weighted branches + multiplicity charge $\implies$ simultaneous concentration,
 simultaneous calibrated intervals $\implies$ certified improvement over NULL,
 future-blind features + uniform value error $\implies$ bounded plug-in regret.

None of the right-hand statements establishes a left-hand premise. In particular, low empirical regret cannot prove causal timing or replay validity.

Result	Nonnegotiable premise	Minimum implementation receipt	Present status
Event timing, Prop. 3.1	Ordered eventizer; branch-local predecessor; valid complete row	Predecessor/successor and candidate digests per event	UNVALIDATED
Sequential support, Thm. 3.2	One-row atomic commit; exact identity NULL	Slot diff and byte equality audit	Draft contract only
Weighted concentration, Thm. 6.1	Bounded fixed loss; conditional independent branches; fixed weights	Branch provenance, bounds, weights, action count	UNVALIDATED
Stable empirical action, Cor. 6.2	Unique gap above simultaneous radius	Per-action estimates and interval separation	UNVALIDATED
Certified NULL, Thm. 7.1	Simultaneous target-law coverage; frozen margin	Final group-level coverage and commit log	UNVALIDATED
Support impossibility, Thm. 8.1	Zero support on positive-mass set	Feasible-set and replay completeness manifest	Conditional theorem
Shift sensitivity, Thm. 8.2	Declared branch laws; bounded loss; valid distance bound	Generator/deployment provenance and shift estimate	No bound established
Value regret, Thm. 9.1	Future-blind inputs; simultaneous uniform error	Runtime field-access audit and per-action error envelope	No model exists
Replay count, Prop. 11.1	Full Cartesian replay	Trace-derived logical and physical counters	No run exists
Pollution bound, Prop. 12.6	Valid per-write conditional error bound	Long-horizon event and repair ledger	No bound established

Table 8: Theorem-to-implementation obligation summary. Appendix C expands the test and failure action for each premise.

13.2 Serialized qualification ladder

1. **Semantic substrate.** Independently qualify a same-checkpoint external semantic row interface under its own registered contract. The frozen evidence does not satisfy this step.
2. **Mechanical fixed-payload oracle.** With one event, exact NULL, complete feasible actions, and grouped futures, price every action without learning content or route.
3. **Oracle qualification.** Apply Protocol 6.3 on fresh prefix groups and required seeds. No reproducible material gap means stop.
4. **Future-blind value.** Only after Step 3, fit Q_ψ and audit leakage, calibration, complete-set regret, NULL behavior, and subgroup stability.
5. **Conservative selector.** Freeze Eq. 33; verify simultaneous coverage and realized improvement without threshold tuning on final groups.
6. **Repeated writes.** Evaluate state-distribution shift, pollution accumulation, repair, archive/index interactions, and lifecycle cost over registered horizons.

Every step produces a versioned manifest and can terminate the line. The ladder does not imply that any step will pass.

13.3 Current boundary

UNVALIDATED The evidence boundary remains before Step 1: the inherited semantic interfaces are failed/unqualified, the fitted-population diagnostic is a fit/support failure for the tested pure latent-row interface, and predictive overwrite remains closed. Steps 2–6 are theory and planned protocol only.

14 Related Foundations, Limitations, and Conclusion

14.1 Related foundations

The MMLA architecture motivates a bounded reasoning-time state, explicit receipts, and a failed/unvalidated semantic-interface boundary rather than supplying evidence for predictive admission [13]. Two companion focus papers provide a typed separation between policy and memory state and an atomic external row contract [14, 12]; they are formal dependencies, not empirical authorities.

Our grouped-future construction uses the potential-outcome distinction between treatment assignment and outcomes [11] and the role of propensity/support conditions in observational adjustment [10]. When logged propensities replace complete controlled replay, contextual-bandit evaluation requires explicit support and estimator assumptions [3]; this does not justify reusing one observed continuation as every action’s counterfactual.

Finite-action confidence control draws on classical bounded concentration [6] and empirical-Bernstein refinements [7]. NULL is related to classification with a reject option [1], but here rejection is also a physical identity transition whose future risk must be replayed. Distributionally robust optimization can expose sensitivity to a declared ambiguity set [2], while conditional value-at-risk supplies one possible tail-risk scalarization [9]; neither method proves that the declared set or loss matches deployment.

Neural Turing Machines, Differentiable Neural Computers, and compressive memory study learned read/write or compressed recurrent memory mechanisms [4, 5, 8]. Predictive admission here concerns a narrower authoritative decision: whether a complete target-conditioned row should replace bounded resident state under a causal future-risk protocol. It neither claims superiority to those systems nor treats their mechanism results as evidence for MMLA.

14.2 Limitations and explicit nonclaims

The theory assumes a finite feasible action set, bounded registered loss, typed complete rows, and a replay system capable of restoring action-conditioned states. Large or continuous action spaces require approximation and new support guarantees. Conditional exchangeability can fail; synthetic futures can be biased; evaluator loss can disorder true outcomes; the deployment law can drift; and a finite horizon can miss slow pollution or repair. The simultaneous bounds can be conservative when actions are numerous or groups are small. A complete teacher can be computationally prohibitive even when deployment is cheap.

No theorem establishes that semantic row content can be decoded at the same checkpoint, that future value is predictable from causal features, that calibrated neural intervals will be obtained, that a safe system exists, or that total-system cost improves. “Certified” in Theorem 7.1 is always relative to stated interval coverage, target law, loss, horizon, and margin. A violated premise voids the deduction.

The manuscript reports no new data collection, experiment, trained model, oracle gap, learned route, learned content, learned shortlist, learned policy, deployment, safety result, or efficiency result. It does not reinterpret the complete mechanical oracle in the fitted-population diagnostic as evidence for learned memory.

14.3 Conclusion

Predictive memory admission should begin with an identified decision problem, not a controller. This paper has separated exact NULL, complete target-conditioned actions, a current-risk comparator, a training-only expected-risk oracle, and a future-blind value estimator. It has made event timing, grouped futures, complete action-conditioned replay, simultaneous uncertainty, branch shift, evaluator error, lifecycle pollution, and the full cost multiplier visible. The resulting ladder is deliberately easy to stop: a failed semantic substrate, absent oracle gap, unpredictable value, uncalibrated selection, long-run pollution, or erased system-level benefit ends progression without relabeling the failure.

CONJECTURED Selective admission may eventually help a qualified bounded authoritative memory. That conjecture remains open. The current scientific result is a formal target, conditional deductions, falsification protocol, and explicit stop boundary.

Author contributions

Junyi Zou and Avrova Donz contributed equally to conceptualization, formal analysis, methodology, visualization design, and manuscript preparation.

Generative AI disclosure

Generative AI tools assisted drafting, mathematical consistency checks, code generation for vector figures, and language editing under human direction. The named authors retain responsibility for claims, sources, and the final manuscript. No AI-generated statement is treated as evidence.

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A Supplementary Proofs and Boundary Lemmas

This appendix expands deductions that separate attainable expected-risk decisions from information ceilings. Every result is conditional on its displayed premises and carries no empirical status.

A.1 Hindsight and expected-risk decisions

PROVED UNDER STATED ASSUMPTIONS

Lemma A.1 (Expectation of the minimum is an information ceiling). *For any finite action set and integrable losses $L(a, F)$,*

$$\mathbb{E} \left[\min_{a \in \mathcal{A}} L(a, F) \right] \leq \min_{a \in \mathcal{A}} \mathbb{E}[L(a, F)]. \quad (54)$$

Equality holds if one fixed action minimizes $L(a, F)$ almost surely, but equality need not imply a unique common minimizer.

Proof. For every fixed $b \in \mathcal{A}$, $\min_a L(a, F) \leq L(b, F)$ pointwise. Taking expectations and then minimizing the right-hand side over b proves Eq. 54. A fixed almost-sure minimizer gives equality. With ties on positive-probability sets, equality can also occur without uniqueness. $\square$

The left side lets the action depend on the realized future and corresponds to Eq. 26. The right side selects one action before the future and corresponds to expected-risk admission. Their difference is the value of prohibited future information, not necessarily a learnable opportunity.

PROVED UNDER STATED ASSUMPTIONS

Proposition A.2 (Current-state loss does not identify future-risk order). *Suppose two actions have fixed observed current costs $C^{\text{cur}}(a)$ and $C^{\text{cur}}(b)$. Without a restriction linking those costs to continuation loss, the sign of $R(a) - R(b)$ is not identified.*

Proof. Construct two continuation laws with identical prefix observations and current costs. In the first, set bounded future losses to $L(a, F) = 0$ and $L(b, F) = 1$; in the second reverse them. The observed current record is identical, while the future-risk ordering reverses. $\square$

A.2 Paired estimation

For independent branches with uniform weights, define $D_k(a, b) = L(a, F_k) - L(b, F_k) \in [-L_{\max}, L_{\max}]$ and $\widehat{\Delta}(a, b) = K^{-1} \sum_k D_k(a, b)$.

PROVED UNDER STATED ASSUMPTIONS

Lemma A.3 (Paired finite-sample bound). *For one preregistered pair (a, b) ,*

$$\mathbb{P} \left(\left| \widehat{\Delta}(a, b) - \Delta(a, b) \right| > \epsilon \right) \leq 2 \exp \left(-\frac{K \epsilon^2}{2L_{\max}^2} \right). \quad (55)$$

For a finite registered set of P pairs, replacing the right side by $2P \exp[-K \epsilon^2 / (2L_{\max}^2)]$ gives simultaneous control.

Proof. Each difference has range length $2L_{\max}$. Hoeffding's inequality for the sample mean gives Eq. 55; a union bound gives the simultaneous statement [6]. $\square$

Pairing does not always improve the worst-case range bound, but Eq. 23 can substantially reduce variance-sensitive intervals. Branches remain the units within one prefix; prefix groups remain the units for population inference.

A.3 Decomposed regret under three error sources

Let R_p^* be target-law risk under target evaluator, R_Q^* proposal-law risk under target evaluator, and $\widetilde{R}_Q$ proposal-law risk under the implemented evaluator. Suppose

$$\sup_a |R_p^*(a) - R_Q^*(a)| \leq \epsilon_{\text{shift}}, \quad (56)$$

$$\sup_a |R_Q^*(a) - \widetilde{R}_Q(a)| \leq \epsilon_{\text{eval}}, \quad (57)$$

$$\sup_a |\widehat{Q}(a) - \widetilde{R}_Q(a)| \leq \epsilon_{\text{model}}. \quad (58)$$

PROVED UNDER STATED ASSUMPTIONS

Theorem A.4 (Target regret under additive uniform errors). *If $\widehat{a} \in \arg \min_a \widehat{Q}(a)$ and $a_p^* \in \arg \min_a R_p^*(a)$, then*

$$R_p^*(\widehat{a}) - R_p^*(a_p^*) \leq 2(\epsilon_{\text{shift}} + \epsilon_{\text{eval}} + \epsilon_{\text{model}}). \quad (59)$$

With Theorem 8.2, one may set $\epsilon_{\text{shift}} = L_{\max} \text{TV}(P, Q)$ when its premises hold.

Proof. The three uniform bounds and the triangle inequality give $\sup_a |R_p^*(a) - \widehat{Q}(a)| \leq \epsilon_{\text{shift}} + \epsilon_{\text{eval}} + \epsilon_{\text{model}}$. Apply Theorem 9.1 with this sum. $\square$

The theorem is a sensitivity decomposition. It is unusable as a certificate until all three terms have valid target-domain bounds.

A.4 Adaptive histories and severe events

Let $V_t \in [0, V_{\max}]$ be contamination severity at write t , with $V_t = 0$ when no contamination occurs. If $\mathbb{E}[V_t \mid \mathcal{H}_{t-1}, A_t \neq 0] \leq v_t$, then by the tower property

$$\mathbb{E} \left[\sum_{t=1}^T V_t \right] \leq \sum_{t=1}^T v_t. \quad (60)$$

This permits adaptive action selection but assumes the conditional bound remains valid under the induced history. A bound calibrated only on a static one-write distribution cannot simply be reused after the selector changes the state distribution.

For a severe-event indicator $B_t = \mathbf{1}[V_t \geq v_{\text{crit}}]$ with conditional rate at most q_t , Proposition 12.6 gives $\mathbb{P}(\max_t V_t \geq v_{\text{crit}}) \leq \sum_t q_t$. If repair removes consequences, the lifecycle loss must model duration and repair cost rather than erase the original event.

A.5 Event-tree work

Suppose segment event j has $m_j(\mathbf{a}_{<j})$ feasible actions after branch prefix $\mathbf{a}_{<j}$. The number of leaf action paths is

$$N_{\text{leaf}} = \sum_{a_1 \in \mathcal{A}_1} \sum_{a_2 \in \mathcal{A}_2(a_1)} \cdots \sum_{a_J \in \mathcal{A}_J(\mathbf{a}_{<J})} 1. \quad (61)$$

If every node has exactly m actions, $N_{\text{leaf}} = m^J$. A frozen rollout changes the estimand and reduces branching after the scored event; it is not a free exact optimization. This is why the first causal oracle test fixes a single event.

B Estimator and Evaluation Protocol Specifications

B.1 Within-group risk record

For each prefix group g , freeze the ordered action manifest $\mathcal{A}_g = (0, a_1, \dots, a_{m_g-1})$, future manifest $(F_{g,1}, \dots, F_{g,K_g})$, nonnegative weights $w_{g,k}$ summing to one, horizon H_g , and scalarization before replay. Store the loss tensor coordinates before aggregating them into $L_{g,a,k}$.

The primary within-group estimates are

$$\widehat{R}_g(a) = \sum_k w_{g,k} L_{g,a,k}, \quad (62)$$

$$\widehat{\Delta}_g(a, b) = \sum_k w_{g,k} (L_{g,a,k} - L_{g,b,k}), \quad (63)$$

$$\widehat{a}_g^{\text{ER}} = \min \arg \min_{a \in \mathcal{A}_g} \widehat{R}_g(a), \quad (64)$$

where the outer min represents the frozen canonical tie order. Effective branch count is Eq. 13; it is reported beside raw K_g . Zero-weight branches remain in provenance but do not inflate effective count.

For the initial gate, do not fit a learned nuisance model to fill missing actions: absence of a required replay slice fails action completeness. If importance weighting is later used to translate proposal law Q_g to target law P_g , require absolute continuity, frozen clipped or unclipped weights, overlap diagnostics, and a bias/sensitivity report. Clipping changes the estimand unless its bias is bounded.

B.2 Population estimands

Let fresh physical prefix groups $g = 1, \dots, G$ be independent draws from the registered target group law. Define paired group advantages

$$\widehat{G}_g^{\text{cur}} = \widehat{R}_g(a_g^{\text{cur}}) - \widehat{R}_g(\widehat{a}_g^{\text{ER}}), \quad (65)$$

$$\widehat{G}_g^0 = \widehat{R}_g(0) - \widehat{R}_g(\widehat{a}_g^{\text{ER}}). \quad (66)$$

Because choosing $\widehat{a}_g^{\text{ER}}$ on the same branches is optimistically biased for the minimum, the registered gate must account for within-group action selection. Acceptable options include a finite-class simultaneous correction, nested branch split with sufficient power, or a bias-corrected procedure fixed in advance. Reporting the raw selected minimum as if it were unbiased is prohibited.

The population aggregate is a fixed weighted mean

$$\widehat{G}^b = \sum_{g=1}^G \nu_g \widehat{G}_g^b, \quad b \in \{\text{cur}, 0\}, \quad \nu_g \geq 0, \text{quad} \sum_g \nu_g = 1. \quad (67)$$

Primary inference resamples or concentrates at the physical-group level. Treating $GK_g m_g$ replay rows as independent is pseudoreplication.

B.3 A conservative finite-class interval

When $0 \leq L \leq L_{\max}$ and weighted branches satisfy Assumption 4.2, use Eq. 29 for a transparent initial interval:

$$[L_g(a), U_g(a)] = \left[\max\{0, \widehat{R}_g(a) - \varepsilon_g\}, \min\{L_{\max}, \widehat{R}_g(a) + \varepsilon_g\} \right]. \quad (68)$$

If actions share branches, preregistered paired empirical-Bernstein intervals can be more efficient [7]; action and pair multiplicity still must be charged. The first result is reported under the simpler registered primary procedure, with alternatives as sensitivity analyses.

B.4 Group bootstrap specification

A group bootstrap draw samples physical prefix groups with replacement and carries each selected group’s full action-by-future tensor. Within-group branches are not independently resampled in the primary population bootstrap. If future-generation uncertainty is also targeted, use a nested bootstrap and report its separate estimand. Stratification variables and weights are frozen before final data are opened.

For B registered bootstrap replicates, the interval construction, quantile convention, studentization, and seed list are fixed. A replicate that lacks a required subgroup is invalid rather than silently dropped. The analysis logs all invalid replicate counts and applies the preregistered failure rule.

B.5 Oracle-gap run sequence

1. Verify the semantic-interface prerequisite, lock the snapshot namespace, and hash every source artifact.
2. Eventize one completed segment into exactly one scored event; freeze fixed payload and complete target actions plus NULL.
3. Construct and validate every complete candidate before future outcomes are exposed to the selector or analyst.
4. Generate or collect grouped futures with provenance; freeze branch weights and target-law claim.
5. Restore the same pre-action snapshot for every action-branch pair and replay the full registered horizon.
6. Compute all loss coordinates, resource counters, per-action risks, paired gaps, and simultaneous intervals automatically.
7. Run action completeness, restoration, future-blindness, group-isolation, evaluator, negative-control, and cost audits.
8. Evaluate fresh prefix groups and all registered seeds once; apply Protocol 6.3 without threshold revision.
9. If any gate fails, record the stop. Only a passed oracle gate permits construction of a separate value-model dataset.

B.6 Value-model evaluation sequence

If and only if the oracle gate passes, fit on training prefix groups, choose all architecture and calibration decisions on calibration groups, freeze the artifact, and evaluate once on final groups. The final table includes the complete action set, per-group regret, raw risk coordinates, NULL confusion metrics, simultaneous coverage, interval width, ambiguity rate, subgroup stability,

negative controls, and full cost. A second final-set pass after model or threshold revision creates a new registered revision and a new untouched final namespace.

UNVALIDATED This appendix is a specification only. It contains no run identifier, data row, interval, effect estimate, or outcome.

C Complete Theorem-to-Implementation Matrix

Table 9: Implementation obligations. “Stop” means the corresponding conclusion cannot be claimed; it does not prescribe a scientific workaround.

Object/result	Premise	Receipt/test	Failure consequence	Evidence status
Completed segment	Immutable decision boundary and ordered bytes	Source digest, cutoff time, tokenizer/build version	Event inputs are ambiguous; stop	Contract only
Eventizer	Deterministic ordered event list under frozen version	Event list digest; rerun equality; human audit sample	Timing theorem inapplicable	Unvalidated
Complete row	Trusted assembly; explicit target/version/protection; parse validity	Byte digest, schema validation, canonical round-trip	Remove action; no semantic claim	Draft dependency
Exact NULL	Identity on resident state, indexes, and authority	Byte-for-byte state and index equality; zero commit trace	Baseline invalid; stop oracle test	Draft dependency
Feasible set	Every valid target represented before outcome access	Deterministic action manifest and rejection reasons	Unsupported-action theorem applies	Unvalidated
Event recursion	Candidate predecessor equals branch prefix	Per-event predecessor/candidate/successor digest chain	Reject branch	Unvalidated
Grouped future	One physical prefix; registered target/proposal law	Group ID closure, timestamps, generator/config hashes	No target-risk interpretation	Unvalidated
Exchangeability	Conditional independent or valid weighted branch draws	Sampling design and dependence diagnostics	Concentration interval invalid	Assumption only
Future causality	Common future exogenous to action, or interventional branch law	Simulator causal contract or domain argument	Counterfactual replay invalid	Assumption only
Snapshot restore	Identical state, cache, index, and RNG boundary	Restoration digest before every action-branch run	Invalidate replay tensor	Unvalidated
Action replay	Memory action can affect hidden evolution and all effects are rerun	Full trace lineage; no pre-action hidden-trace reuse	Horizon loss not action conditioned	Unvalidated
Loss bound	Every scalar loss lies in registered $[0, L_{\max}]$	Automated range audit and overflow log	Hoeffding claims void	Unvalidated
Loss meaning	Evaluator linked to target task/truth	Frozen evaluator, blinded validation, sensitivity envelope	Only proxy-risk claim remains	Unvalidated
Current comparator	No realized future; definition and ties frozen	Comparator artifact hash and runtime field log	Gap comparison invalid	Unvalidated
Empirical ER oracle	Complete actions, valid risks, multiplicity correction	Action tensor and simultaneous intervals	No oracle-gap claim	Unrun
Population gap	Fresh independent prefix groups; registered weights/subgroups	Final namespace manifest and group-level inference	No population promotion	Unrun
Future-blind features	Every field pre-action measurable	Static dataflow graph plus runtime access log	Reject model as leaked	No model
Value error	Simultaneous complete-action error bound on target groups	Per-action prediction/risk envelope	2ϵ theorem unusable	No model
Calibrated selector	Simultaneous coverage and frozen margin	Final coverage, interval, action, and commit receipts	NULL-only fallback	No model
Branch shift	Target/proposal distance bounded in relevant loss class	Declared law and sensitivity bound	Synthetic gap cannot transfer	No bound
Repeated writes	One-write guarantees stable under induced histories	Stateful multi-event replay and pollution/repair ledger	No lifecycle claim	Unrun
Complete cost	Every constructor, replay, archive, index, and deployment term counted	Logical counts, physical counters, reconciliation audit	No efficiency claim	Unrun

The matrix is deliberately redundant with the prose. A theorem citation without the corresponding receipt is an incomplete claim. Receipts must identify the exact artifact revision; evidence from another checkpoint, action set, horizon, evaluator, or branch law cannot be silently reused.

D Symbol, Status, and Boundary Ledger

D.1 Core symbols

Table 10: Primary notation. All group-indexed objects inherit the registered prefix group g .

Symbol	Type	Meaning
M_{n-1}	bounded resident state	Authoritative state before completed segment n

Symbol	Type	Meaning
x_n	completed segment	Immutable observation block at a decision boundary
$e_{n,j}$	typed event	Ordered event j extracted from x_n
$M_{n,j}$	bounded resident state	State after decisions through event j
$c_{n,j,i}$	complete row	Target-conditioned candidate for action i
$\mathcal{A}_{n,j}$	finite action set	Exact NULL plus all valid complete row commits
0 or NULL	action	Physical identity transition; no row or authority change
$\tilde{S}_g$	causal prefix record	Typed pre-action state for prefix group g
$\mathcal{D}_g$	sigma-field	Information legally available to deployment
$F_{g,k}$	future branch	Training/evaluation-only continuation k sharing prefix g
Q_g, P_g	probability laws	Proposal/grouped-future law and target deployment law
$w_{g,k}$	scalar	Nonnegative branch weight summing to one
K_g^{eff}	scalar	Effective weighted branch count $(\sum_k w_{g,k}^2)^{-1}$
$Y_{g,a,k,h}$	replay outcome	Action-conditioned outcome at horizon cell (a, k, h)
$L_g(a, F)$	bounded scalar	Registered horizon loss after action a under future F
$\tilde{R}_g(a)$	estimator	Weighted empirical action risk under grouped branches
$R_{Q_g}(a)$	estimand	Conditional risk under proposal law
$R_{P_g}(a)$	estimand	Conditional risk under target deployment law
a_g^{cur}	comparator action	Frozen current-risk minimizer without realized future
$\tilde{a}_g^{\text{ER}}$	teacher action	Empirical expected-risk minimizer over complete actions
$a_{g,h}^{\text{hind}}$	information ceiling	Per-realized-future minimizer; unavailable at deployment
G_g^{cur}, G_g^0	risk differences	Oracle advantages over current comparator and NULL
$X_{g,a}$	typed features	Future-blind deployment features for one action
$Q_\psi(X_{g,a})$	model output	Predicted conditional action risk; no model exists here
$[L_g(a), U_g(a)]$	interval	Simultaneous risk interval for action a
μ	fixed scalar	Materiality/commit margin
Reg_g	nonnegative scalar	Excess target risk of selected versus best action
$\mathfrak{R}$	resource vector	Tokens, compute, bytes, traffic, latency, energy, and storage terms

D.2 Literal status vocabulary

Printed marker	Interpretation in this manuscript
DEFINITION	Object or estimand defined; no existence, quality, or implementation claim
PROVED UNDER STATED ASSUMPTIONS	Deduction valid when every displayed premise holds; premises not empirically established by the proof
CONJECTURED	Open falsifiable hypothesis with no admitted supporting result
DESIGN PRESCRIPTION	Required design or receipt for a conforming future test
PLANNED TEST	Registered-form test described but not executed
UNVALIDATED	Gate failed, unopened, or unsupported by accepted evidence

Table 11: Status markers are local. A proved lemma cannot promote the system that may someday instantiate it.

D.3 Frozen evidence boundary

- PM-I2: 0/9 jobs qualified.
- PM-I3 learned-content exactness: 0/73,728 for learned routing/learned content and 0/73,728 for oracle routing/learned content.
- PM-I3 learned routing with oracle content: 18,544/73,728; this does not establish learned content.
- PM-I3 complete mechanical oracle: 73,728/73,728; this is a mechanical ceiling, not evidence for a learned route, learned content, or predictive policy.
- Accepted terminal: fit/support failure for the tested pure latent-row interface; it does not isolate representation, decoder, or optimization and does not reject every possible MMLA realization.
- Predictive admission: no executed oracle test, no measured gap, no trained value model, and no learned policy.

These values are inherited only to delimit what this theory paper may claim. They are not reanalyzed, improved, or converted into a predictive-admission result.